

Sierra Manriquez, M18532809

MEEM 2911, L14, Spring 2025

Cooper Evans

MAEP 2 Reflective Essay

MAEP II has been an effective way to apply concepts from past engineering courses to real world projects. This course's main project is adapting air handling unit to meet flow requirements for a mobile Neonatal Intensive Care Unit. I chose three assignments to include in this portfolio because they represent how much I have learned. In the Week 5 team assignment on dynamic temperature measurement, I learned how to design a test rig using cost effective materials like a hair dryer to simulate a heater, and analyze sensors based on accuracy, cost, and response time. I also attended an extra, optional lab session independently to write a detailed standard operating procedure. In the Week 7 team assignment, I learned flow measurement principles and how to analyze uncertainty by comparing orifice plates to ASME standards and then made design recommendations based on uncertainty and flow rates. The Week 10 individual assignment increased my skills of applying mechanics of materials to design an orifice plate, including a CAD model and engineering drawing. For all assignments of MAEP II, I consistently used feedback to strengthen my understanding.

The most challenging part of MAEP II was writing a MATLAB script that used an iterative guessing approach to calculate flow properties such as Reynolds number and discharge coefficient. Not only did this require programming skills, but it also required a strong understanding of the governing equations and when to apply specific, special case conditions defined in the ASME Standard. Translating engineering equations from text to functional code was difficult, especially when creating conditional statements to determine when to account for any special cases. Every day, I reviewed lecture material and reread chapters in the Reference Articles to strengthen my understanding. I also applied knowledge from past courses such as thermodynamics and material science, which were critical for understanding equations and results.

I have greatly developed my ability to design and analyze engineering systems, putting emphasis on measurement accuracy. I gained hands on project experience, selecting, designing, and validating a flow measurement device, ultimately choosing an orifice plate because it balanced accuracy and manufacturability. I am confident that I can apply standards, perform uncertainty analysis, and use CAD to create detailed engineering drawings with necessary information. I increased my ability to interpret experimental data and then make calculated design choices based on trade-offs and balancing cost, accuracy, and performance. I expect to apply these skills in my engineering career, especially in roles that involve testing, validation, and system design.

Communicating technical information is a skill that I have improved significantly during MAEP II. When I compare my current work to previous reports, I now understand the importance of proper formatting, clarity, and data presentation because it influences the readers' understanding. Weekly visits to my GTA's office hours played a big part in this

improvement because I was able to receive direct feedback and apply it to upcoming assignments. I also communicated the GTA feedback to my teammates to make sure they were up to date with information and to ensure all our team submissions were consistent. Specifically, the Week 7 team assignment required the team to evaluate each team members' orifice plate design and then justify a final selection. This required me to clearly explain to my teammates why certain designs were not chosen while being constructive and professional about communicating. In the future, I will continue to improve my communication skills by practicing concise technical writing and seeking feedback.

The biggest challenge I faced with teamwork was initially finding my role within the group. Early into the semester, roles were not clearly defined which made collaborating on assignments less efficient. Over time, I took on the role of meeting with the GTA to review assignments and then relay feedback to the team. I ensured that assignments were corrected before resubmission. Compared to MAEP I, where my role was more of a project leader, my role in this course was more supportive and adaptive. I often took on tasks that others were less comfortable with, and in many cases, I needed to learn new methods to complete them. This experience has taught me the importance of being flexible for others. My advice to incoming students is to work hard, find passion in what you do, and step outside your comfort zone to support your team's success.



**Michigan
Technological**
University

Introduction to Dynamic Measurements of Temperature

Declan Jordan - 25% Sierra Manriquez - 25% John Dombrowski - 25% Andrew Shepard - 25%

L14 Team 2

Temperature Responses and Measurement

1

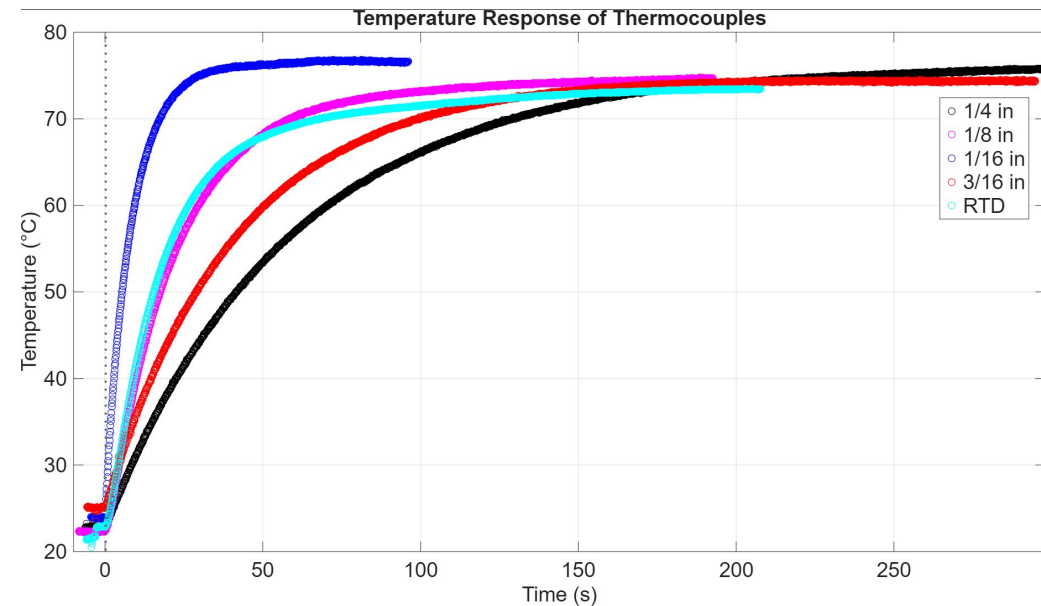


Figure 1. Temperature Vs. Time graph for Each Thermocouple

Figure 1 shows a graph of the data from each thermocouple as well as the existing 1/8 inch RTD sensor. This shows the relative speed of the smaller sensors, with the time constant being inversely proportional in some way to the size of the sensor. Only the 1/16 inch and 1/8 inch thermocouples perform better than the RTD sensor based solely on this graph.

Table 1 shows the maximum and minimum readings from each sensor, corresponding to the final temperature and the temperature where the trial starts respectively. The minimum temperature is shown on the graph at time equals 0, with everything prior being before the trial begins.

Table 1. Significant Values from Each Thermocouple

	1/4 in	1/8 in	1/16 in	3/16 in	RTD
Max Temp (°C)	75.82	74.77	76.83	74.51	73.48
Min Temp (°C)	22.85	22.43	23.94	25.03	22.88

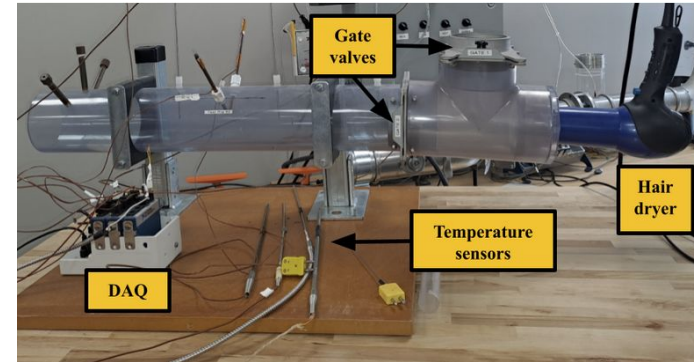


Figure 2. Annotated Test Rig

The data was collected using the test rig shown in Figure 2, with the temperature sensor highlighted in Figure 3. This rig is constructed with a hair dryer blowing warm air through valves to control temperature and airflow. This allowed for fine temperature control and repeatability of the trial for each sensor.

The temperature sensors were connected to a DAQ that saved the data from each trial to be exported to MATLAB for analysis.

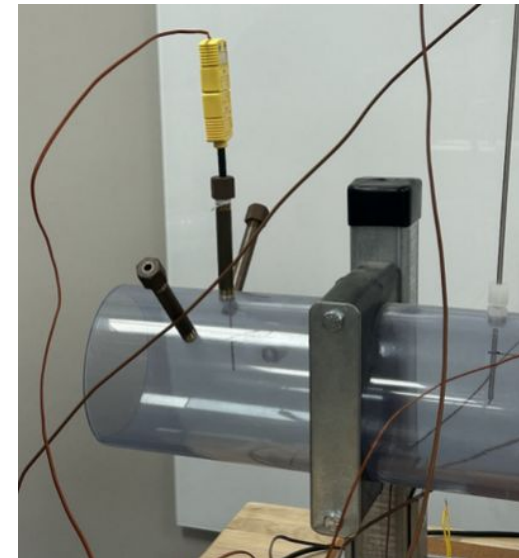


Figure 3. Temperature Sensor in Test Rig

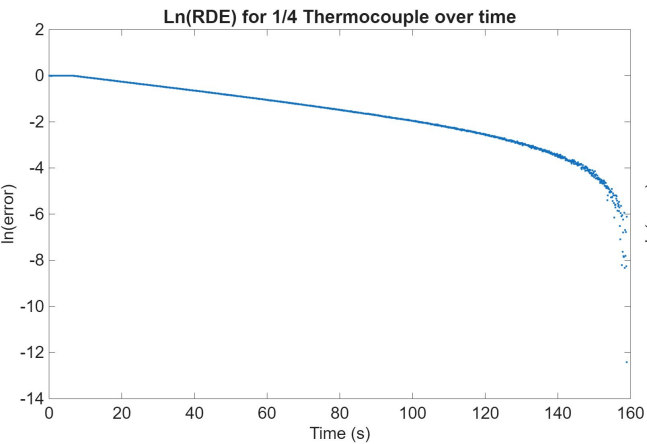


Figure 4. $\ln(\text{Relative Dynamic Error})$ 1/4"

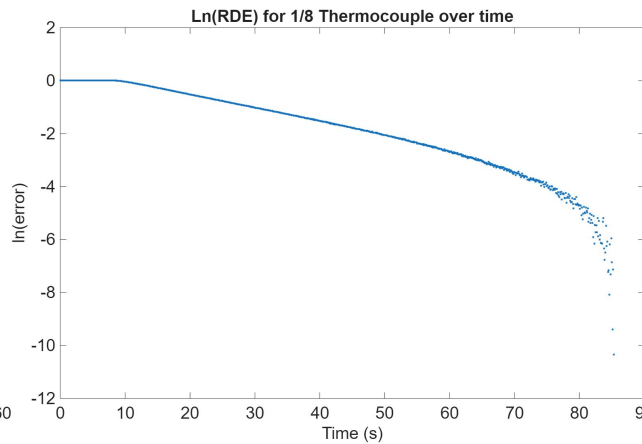


Figure 5. $\ln(\text{Relative Dynamic Error})$ 1/8"

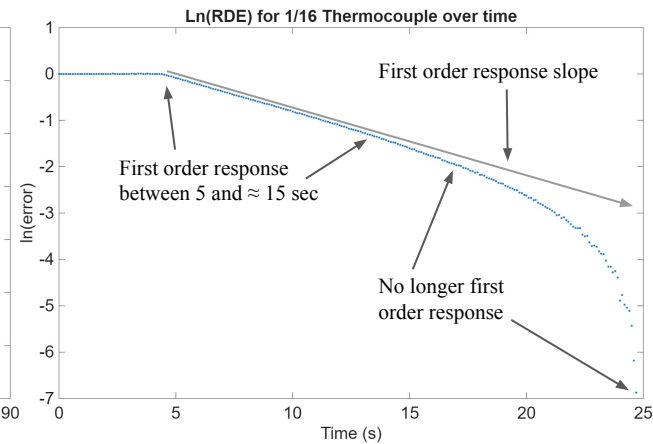


Figure 6. $\ln(\text{Relative Dynamic Error})$ 1/16"

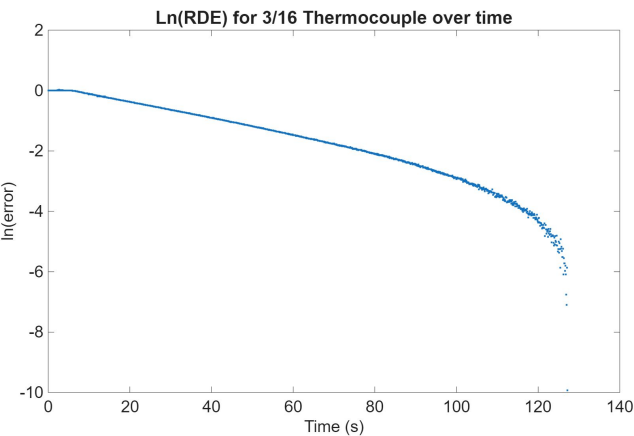


Figure 7. $\ln(\text{Relative Dynamic Error})$ 3/16"

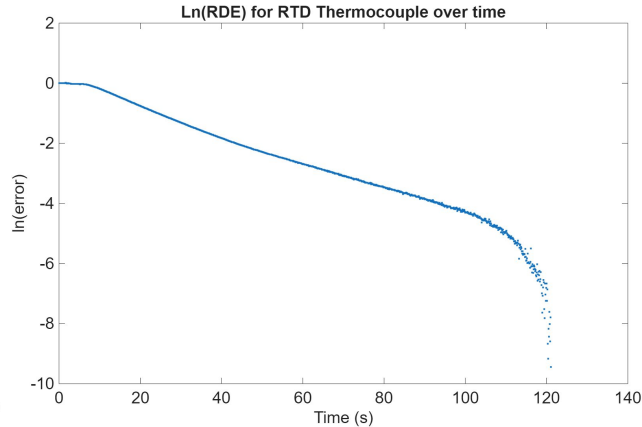


Figure 8. $\ln(\text{Relative Dynamic Error})$ RTD

Relative dynamic error measures how far away a system is from its final target value. As seen from the plots the thermocouples readings tend to take some time till they reach steady state. This is most likely due to thermal inertia, causing the thermocouple to take some time to provide a proper reading. This is also most likely the reason for both first and non first order responses. This is because when there is a large difference in temperature between the rod and its surroundings, heat transfers as quickly as the material will allow, and as the thermocouples temperature approaches its surroundings temperature, heat transfer gradually becomes slower and slower due to diminishing temperature gradients, occurring when two temperatures approach equilibrium.

1st-Order Natural log of Relative Dynamic Error Over Time Comparison

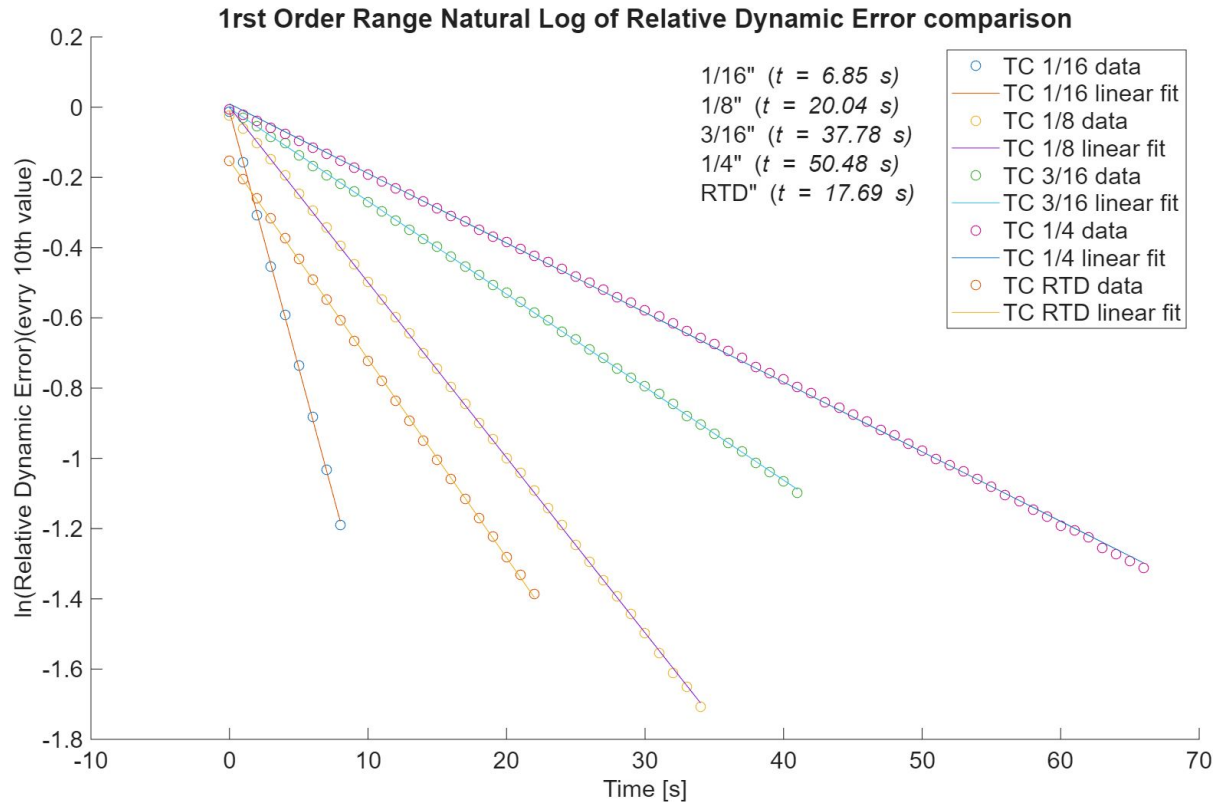


Figure 9. Comparison of first order response of $\ln(\text{RDE})$ over time using every tenth value

Table 2. Thermocouple slopes and time Constants

Thermocouple type	Slope (m)	Time Constant (sec)
1/16	-0.1460	6.85
1/8	-0.0499	20.04
3/16	-0.0265	37.78
1/4	-0.0198	50.48
RTD	-0.0565	17.69

To be able to calculate the sensor's time constraints, the slopes of their first order portion need to be found first. To do this a line of best fit was applied to each sensor's first order range using matlab and their slopes were taken. Once the slopes are found all that's left is to divide -1 by the slope of each sensor to get their time constants.

The required supply air temperature is $23 \pm 1^\circ\text{C}$; therefore, any replacement sensor must provide accuracy within $\pm 1^\circ\text{C}$ while responding fast enough for feedback control. According to Omega, standard Type-K thermocouples have limits of error of $\pm 2.2^\circ\text{C}$ or $\pm 0.75\%$ of reading (whichever is greater). At operating temperatures near $70\text{--}80^\circ\text{C}$, this corresponds to $\pm 2.2^\circ\text{C}$, which does not meet the $\pm 1^\circ\text{C}$ requirement. Special Limits of Error (SLE) Type-K thermocouples improve to $\pm 1.1^\circ\text{C}$ or $\pm 0.4\%$, which is slightly above $\pm 1^\circ\text{C}$ and therefore borderline. According to NIST ITS-90 thermocouple reference data, Type-T thermocouples provide tighter limits of error of $\pm 1.0^\circ\text{C}$ or $\pm 0.75\%$, and SLE Type-T improves to approximately $\pm 0.5^\circ\text{C}$, clearly satisfying the $\pm 1^\circ\text{C}$ requirement. Therefore, if strict compliance is required, Type-T is preferred over Type-K.

The duct diameter is 102.36 mm. Instrumentation guidelines recommend probe diameter not exceed approximately 5% of duct diameter: $0.05 \times 102.36 = 5.12$ mm. This corresponds to a maximum practical diameter of approximately 3/16 in. to avoid flow disturbance. However, the system settling time requirement is ≤ 76 s. For a first-order response: Settling time $\approx 5\tau \leq 76 \Rightarrow \tau \leq 15.2$ s. Only the 1/16 in thermocouple ($\tau = 8.34$ s) satisfies this dynamic requirement. All larger diameters exceed the allowable time constant. Thus, while 3/16 in meets geometric constraints, it fails the dynamic requirement. The maximum acceptable diameter based on dynamic response is 1/16 in.

According to Omega product specifications, grounded junction thermocouples provide faster response due to direct thermal contact between the junction and sheath, while ungrounded junctions primarily reduce electrical noise. Because this application prioritizes rapid temperature feedback and no electrical interference concerns were identified, a grounded junction configuration is recommended.

Table 3. Time Constant and Cost Comparison of the Various Sensors

Sensor	Diameter	τ (s)	5τ (s)	Cost (\$)	Accuracy vs $\pm 1^\circ\text{C}$	Meets Requirements
RTD	$\frac{1}{8}$ in	19.42	97	122.40	Excellent ($\pm 0.10^\circ\text{C}$)	Accuracy only
Type-K	$\frac{1}{4}$ in	58.26	291	65.00	No	No
Type-K	3/16 in	41.32	206	55.00	No	No
Type-K	$\frac{1}{8}$ in	23.02	115	94.48	No	No
Type-K (SLE)	1/16 in	8.34	41.7	60.00	Borderline	Time only

Based on accuracy and dynamic response requirements, the recommended replacement for the RTD is a 1/16 in grounded thermocouple. To fully satisfy the $\pm 1^\circ\text{C}$ accuracy requirement, a 1/16 in grounded Type-T thermocouple (preferably Special Limits of Error) is recommended. This configuration: Meets the $\pm 1^\circ\text{C}$ accuracy requirement (per NIST data), Satisfies the ≤ 76 s settling requirement, Minimizes airflow disturbance, Reduces cost by approximately 50% compared to the RTD.

Therefore, a 1/16 in grounded Type-T thermocouple is the most appropriate replacement for the RTD in the mNICU supply air system.

ME 2911 – MAE Practice II

Temperature Sensor Test Rig

Standard Operating Procedures

February 11, 2025

1 Overview	2
2 Safety	2
3 Hair Dryer Operation	2
4 Data Collection	3
5 Testing Considerations	5
6 Troubleshooting Tips	5
7 References	6

1 Overview

The test rig shown in Figure 1 is used to test the dynamic response of different temperature sensors, including four different thermocouples and one RTD. The hair dryer is attached to the right-hand side of the test rig with the purpose of generating hot air flow. Two gate valves are located to the left of the hair dryer with the purpose of directing air flow. Adjusting the gate valves can direct air down the tube towards the temperature sensors or through a bypass that exhausts to the room. The DAQ, located on the left-hand side, measures and collects data from the temperature sensor that is then written into a data file using the LabView interface.

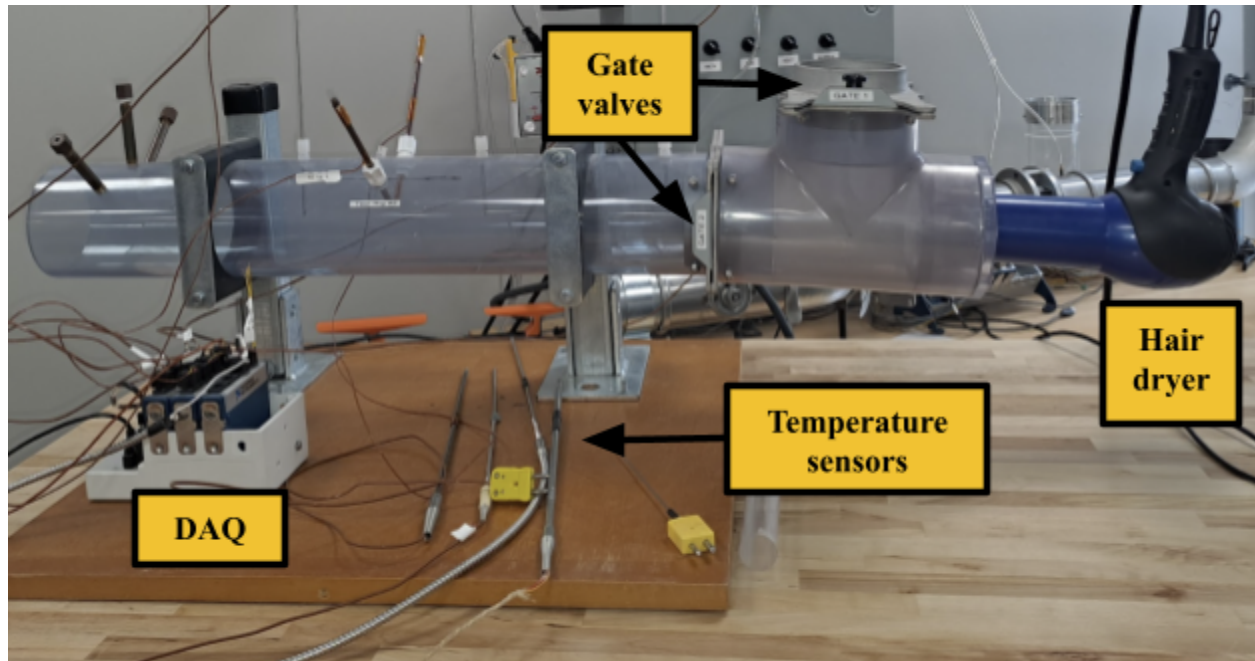


Figure 1. Test rig with annotations showing major components

2 Safety

Safety goggles must be worn at all times while operating the dynamic sensor test rig due to potential hazards. Flying debris and electrical shock are a threat during operation because the test rig uses blowing air and electrical heaters. During operation, neither gate valve should be closed at the same time because the hair dryer will overheat due to a lack of airflow. Additionally, the hardware used for the rig may fail, so it is important to be aware of the surroundings and inspect the hardware for any potential hazards before and after operation.

3 Hair Dryer Operation

There are five operational modes, which are combinations of the hair dryer's settings. The first operational mode is a low temperature setting and a low fan speed setting. The second mode is the low temperature

setting with high fan speed. The third mode is high temperature with low fan speed. The fourth mode is high temperature with high fan speed. These modes are activated by flipping the fan and heat settings switches on the hair dryer. The fifth operational mode is a cool-down, where the heater is off, but the fan still runs. This mode is activated by the blue button. For testing, the fourth operational mode will be used, where the hair dryer runs at high fan speed and high temperature. Figure 2 shows the hair dryer in more detail.



Figure 2. Close-up of hair dryer component of test rig

4 Data Collection

The LabView user interface is located in the C Drive in the folder AHU with the name 'Temperature Main 0.8'. After clicking on the file to open the software, the interface shown in Figure 3 will appear. The legend shows sensor readings of temperature in units of Celsius and voltage in units of millivolts. The acquired Data section shows the sensor readings in real time in the form of a graph.

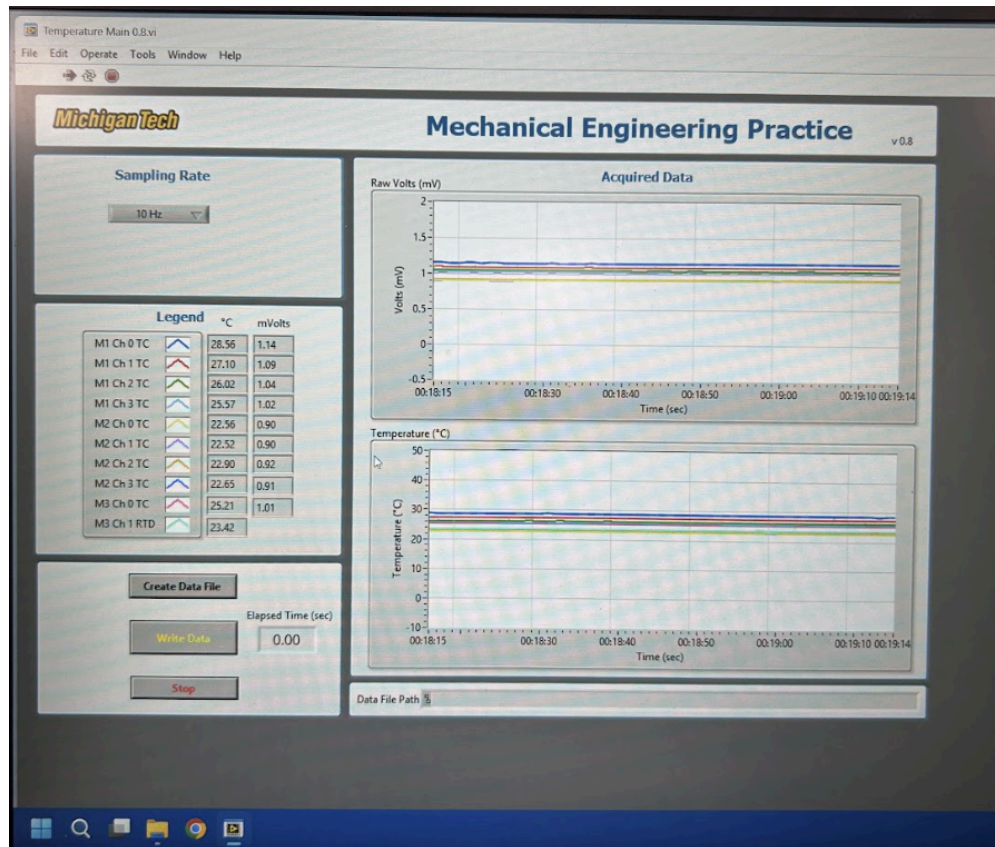


Figure 3. Lab View user interface after opening

To create the data file, click on 'Create Data File', and when prompted, choose to save the data file in a location where it can be easily found so that the data can be plotted for verification. By default, the data will be saved to a folder located in the C Drive. Click on 'Write Data' to start collecting data, and to save the data, click on 'Write Data' again. Clicking on 'Stop' will stop the LabView program [1].

To collect data that will allow for analyzing temperature sensor response, a step change in temperature must be mimicked using hot air flow from the hair dryer. To do this, the temperature sensor must be inserted into the test rig after the sensor is reading a constant room temperature and the hair dryer has had time to heat up. Figure 4 shows a temperature sensor properly inserted into the test rig.

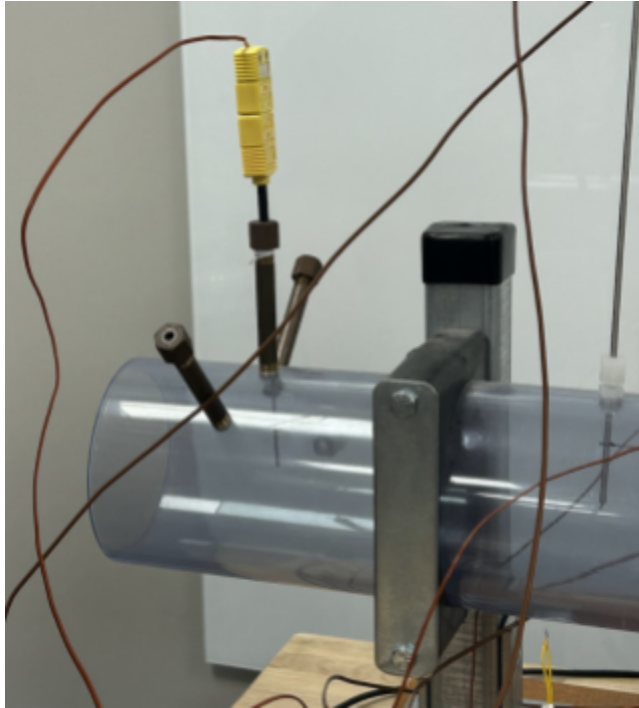


Figure 4. Temperature sensor inserted into test rig

5 Testing Considerations

In an emergency situation, the rig can be shut down by turning off the hair dryer or by unplugging the rig from the wall entirely to stop all flow of electricity through the rig.

Remember that the temperature sensors are sensitive, meaning they will pick up body heat transferred from people nearby and from touching the sensor. This added heat will create a bias in the collected data because the initial temperature reading will not be at room temperature.

The software does not specify which sensor is being recorded so it is important to trace the sensor to the channel it is plugged into on the DAQ. Then, make note of each test run and its corresponding sensor, to eliminate confusion when analyzing data.

6 Troubleshooting Tips

The hair dryer may stop unexpectedly due to thermal overload of the circuit breaker located in the plug. There is a push button, integral to the electrical plug, for resetting the breaker. [2]

Critical data should not be taken immediately. The DAQ may need to warm up for at least 10 minutes before it will be ready to start writing data.

7 References

[1] “ME2911-SOP-TemperatureRig”, in ME2911 canvas course, last accessed on 02/11/2026

[2] “ME2911.W5.TA”, in ME2911 canvas course, last accessed on 02/11/2026

Exercise 1

given: $k_p = 30$, $T_i = 0.65 \text{ min}$

$$k_i = \frac{k_p}{T_i} = \frac{30}{0.65} = 46.1538 \frac{1}{\text{min}} = k_i$$

Figure 1. Calculations for integral gain (1/min)

Exercise 2

2a.

This response is classified as stable and underdamped because, as time approaches infinity, the amplitude reaches a constant value, and there is oscillation.

2b.

The magnitude of overshoot is about 0.025 V.

2c.

The settling time is about 11 seconds.

2d.

The steady-state error is 0.03 V.

Exercise 3

3a.

given: $\sigma_{y,min} = 4496 \text{ Psi}$, $\sigma_{y,max} = 5076 \text{ psi}$

$\sigma_{u,min} = 5366 \text{ Psi}$, $\sigma_{u,max} = 7397 \text{ psi}$

$A = 3.1 \text{ mm} \times 7.8 \text{ mm}$ $A = 0.0031 \times 0.0078 = 2.418 \times 10^{-5} \text{ m}^2$

$$\frac{2.418 \times 10^{-5} \text{ m}^2}{1} \cdot \left(\frac{39.37 \text{ in}}{1 \text{ m}} \right)^2 = 0.0375 \text{ in}^2$$

$F = \sigma A$, $1 \text{ lbf} = 4.44822 \text{ N}$

$F_{y,min} = \sigma_{y,min} A = 4496 \times 0.0375 = 168.5052 \text{ lbf} = \text{minimum yield force}$
 $\frac{168.5052 \text{ lbf}}{1} \cdot \frac{4.44822 \text{ N}}{1 \text{ lbf}} = 749.5484 \text{ N}$

$F_{y,max} = \sigma_{y,max} A = 5076 \times 0.0375 = 190.2430 \text{ lbf} = \text{maximum yield force}$
 $\frac{190.2430 \text{ lbf}}{1} \cdot \frac{4.44822 \text{ N}}{1 \text{ lbf}} = 846.2428 \text{ N}$

$F_{u,min} = 5366 \times 0.0375 = 201.1119 \text{ lbf} = \text{minimum ultimate force}$
 $\frac{201.1119 \text{ lbf}}{1} \cdot \frac{4.44822 \text{ N}}{1 \text{ lbf}} = 894.5900 \text{ N}$

$F_{u,max} = 7397 \times 0.0375 = 277.2316 \text{ lbf} = \text{maximum ultimate force}$
 $\frac{277.2316 \text{ lbf}}{1} \cdot \frac{4.44822 \text{ N}}{1 \text{ lbf}} = 1233.1872 \text{ N}$

Figure 2. Calculations for tensile force at which this component will fail using maximum and minimum values of yield and ultimate stress for PLA.

3b.

$$\text{given: } E_{\min} = 299000 \text{ psi}, E_{\max} = 374000 \text{ psi}$$

$$\epsilon = \frac{\sigma}{E}$$

$$E_{\min} = \frac{\sigma_{y,\min}}{\epsilon_{\max}} = \frac{4496}{374000} = 0.0120 = \text{Minimum possible strain}$$

$$E_{\max} = \frac{\sigma_{y,\max}}{\epsilon_{\min}} = \frac{5076}{299000} = 0.0170 = \text{Maximum possible strain}$$

$$\boxed{\text{Range of possible strain} = 0.0120 - 0.0170}$$

Figure 3. Calculations for range of possible strains expected when component is at yield point

$$\text{given: } L = 9.9 \text{ mm}$$

$$\Delta L = \epsilon L$$

$$\Delta L_{\min} = E_{\min} L = (0.0120)(9.9 \text{ mm}) = 0.1190 \text{ mm}$$

$$\Delta L_{\max} = E_{\max} L = (0.0170)(9.9 \text{ mm}) = 0.1681 \text{ mm}$$

$$\boxed{\text{Range of possible deformations} = 0.1190 - 0.1681 \text{ mm}}$$

Figure 4. Calculations for range of deformations (mm) if the component is 9.9 mm in length

3c.

given: $P = 143 \text{ kPa} = 143000 \text{ Pa}$, $D = 127 \text{ mm} = 0.127 \text{ m}$

$$r = \frac{D}{2} = \frac{0.127}{2} = 0.0635 \text{ m}$$

$$A = \pi r^2 = \pi (0.0635)^2 = 0.0127 \text{ m}^2$$

$$F = PA = (143000)(0.0127) = 1811.4792 \text{ N}$$

$$\frac{1811.4792 \text{ N}}{1} \cdot \frac{0.22480916 \text{ lbf}}{1 \text{ N}} = 407.2368 \text{ lbf} = F$$

Figure 5. Calculations for force that corresponds pressure loaded on cylindrical plate

Exercise 4

4a.

The maximum pressure in kPa that the PLA orifice plate can withstand before failure is the smallest ultimate strength value listed in Table 22.3 of the reference articles. This value is 5366 psi, which converts to 36997.2676 kPa. The smaller end of the range of ultimate stress values is chosen because it is the first possible pressure that the part could fail at.

4b.

From the Matlab code, "Week10_IA", attached to this submission, it was calculated that the highest differential pressure applied to the orifice plate is 1773 Pa.

4c.

From part a, the maximum pressure that the PLA orifice plate can withstand is 36997.2676 kPa. From part b, the highest differential pressure applied to the orifice plate is 1773 Pa. Since the highest differential pressure applied to the orifice plate is less than the maximum pressure that the PLA orifice plate can withstand, the printed PLA orifice plate will not fail under differential pressure when used for flow rate measurement.

Exercise 5.

from DP1: $\beta = 0.5915$, $D = 95 \text{ mm}$

$$\beta = \frac{d}{D} \Rightarrow d = \beta D = (0.5915)(95) = 56.1925 \text{ mm} = d$$

$$\begin{aligned} \delta_{\max} &= \frac{0.005(D-d)}{2} = \frac{0.005(95-56.1925)}{2} \\ &= \frac{0.0970 \text{ mm}}{1} \cdot \frac{1 \text{ mm}}{1000 \text{ mm}} \cdot \frac{10^6 \text{ } \mu\text{m}}{1 \text{ mm}} = \boxed{97.0186 \text{ } \mu\text{m} = \delta_{\max}} \end{aligned}$$

Figure 6. Calculations for maximum allowed deflection of the orifice plate (micrometers) using the deflection equation from the ASME standard

Exercise 6

F	50	degrees
H	102.36	mm
β	0.5915	[-]
d	56.1925	mm
e/D	0.0189	[-]
e	1.8	mm
E/D	0.0379	[-]
E	3.6	mm
W	2.8003	mm
G	60.4828	mm

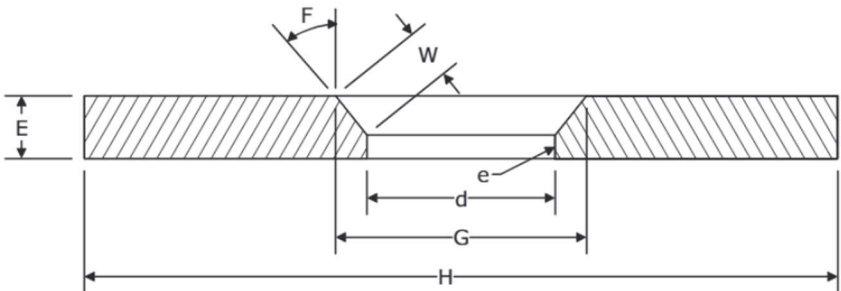


Figure 1. Filled table of necessary dimensions for designing the orifice plate

given: $F = 50^\circ$, $H = 102.36 \text{ mm}$, $\beta = 0.5915$, $D = 95 \text{ mm}$
 $E = 3.6 \text{ mm}$, 3D print layer height = 0.2 mm

$$0.005D < e < 0.02D \Rightarrow 0.475 < e < 1.9$$

$$\frac{e}{\text{layer height}} = \frac{1.8}{0.2} = 9 \quad \checkmark \quad \text{Choose } \boxed{e = 1.8 \text{ mm}} \text{ b/c it is a multiple of layer height}$$



$$\frac{G-d}{2} = W \sin(F)$$

$$E-e = W \cos(F) \Rightarrow W = \frac{E-e}{\cos(F)} = \frac{(3.6) - (1.8)}{\cos(50^\circ)} = \boxed{2.8003 \text{ mm} = W}$$

$$G = d + 2W \sin(F) = 60.5459 + 2(2.8003) \sin(50^\circ)$$

$$\Rightarrow \boxed{G = 60.4828 \text{ mm}}$$

$$B = \frac{d}{D} \Rightarrow d = BD = 0.5915(95) \Rightarrow \boxed{d = 60.5459 \text{ mm}}$$

$$\frac{e}{D} = \frac{1.8}{95} = \boxed{0.0189 = \frac{e}{D}}, \quad \frac{E}{D} = \frac{3.6}{95} = \boxed{0.0379 = \frac{E}{D}}$$

Figure 2. Calculations for unknown dimensions

Exercise 7

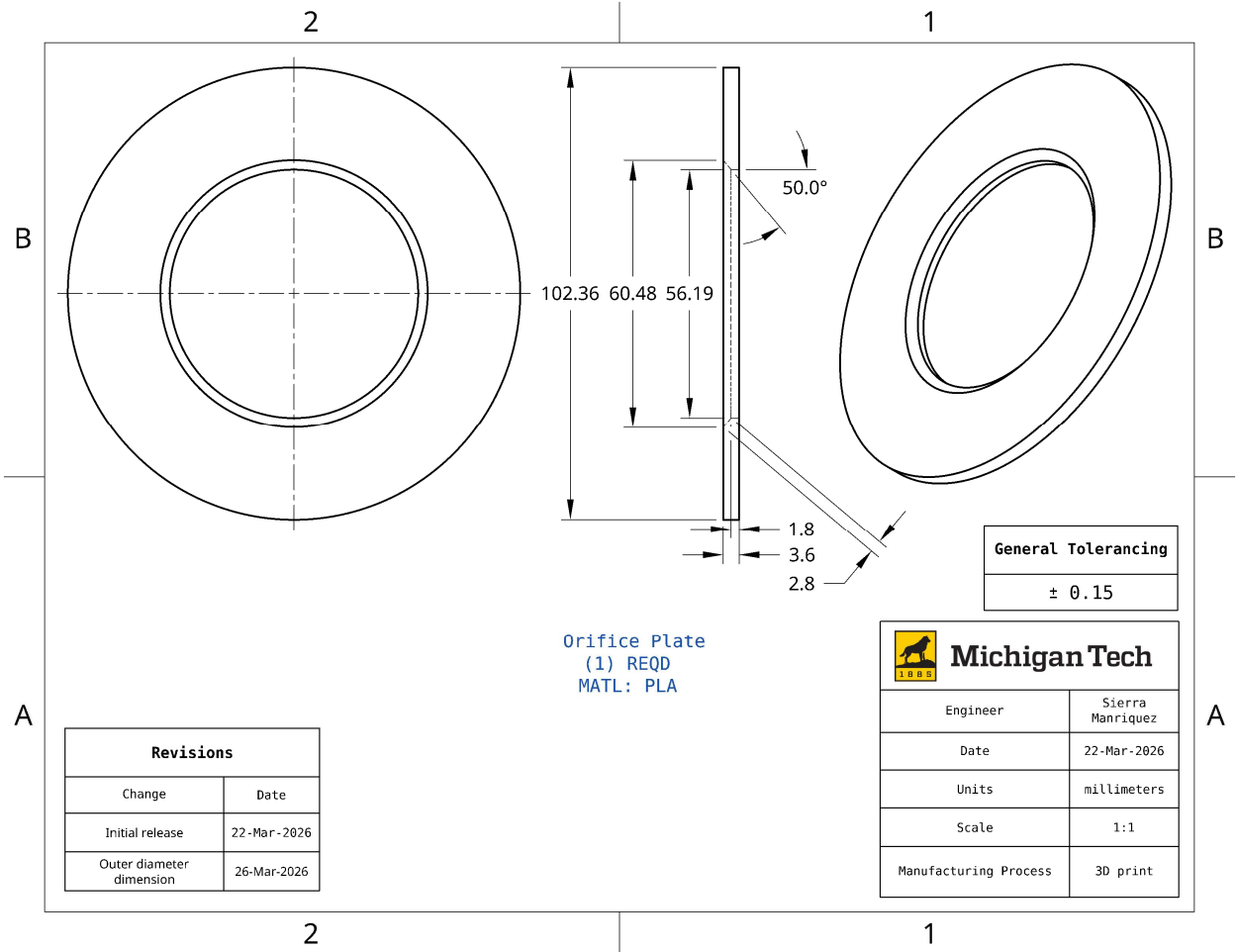


Figure 3. Detailed technical part drawing of the orifice plate



Michigan
Technological
University

Flow Measurement Principles, Performance Uncertainties, and Design Recommendations

Declan Jordan - 25% Sierra Manriquez - 25% John Dombrowski - 25% Andrew Shepard - 25%

L14 Team 2

Introduction to flow measurement principle and design trade-offs

1

Pressure transducer voltage was measured using pressure transducer PT2A. The measured voltage was converted to pressure head in inches of water column using the transducer calibration. Then this data was processed in matlab using iterative guessing with and equations from ASME MFC-3M-2004 to determine discharge coefficient, reynolds number, and mass flow rate.

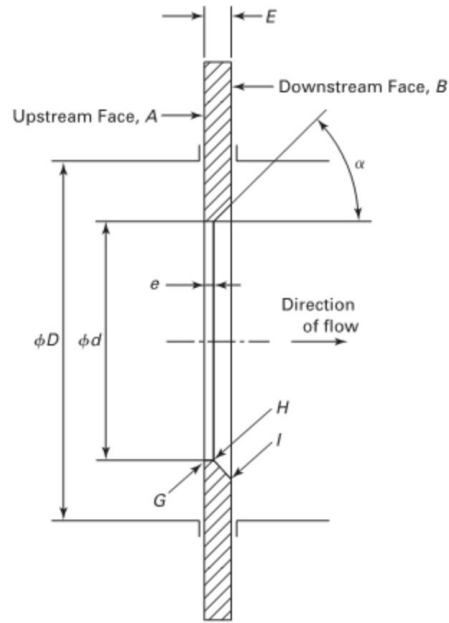


Figure 1. Drawing of standard orifice plate cross section

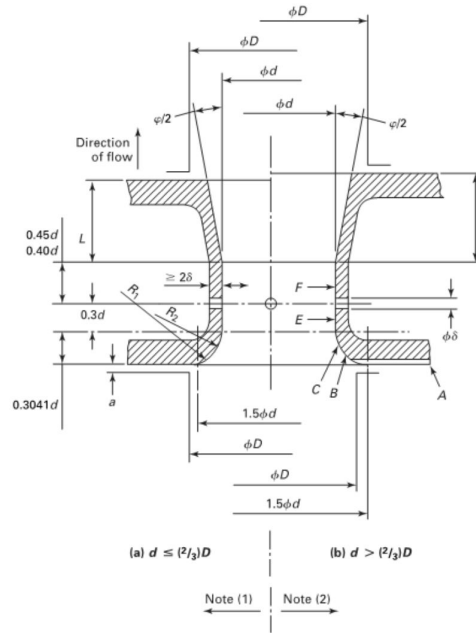


Figure 2. Drawing of venturi nozzle cross section

The orifice plate and nozzle devices shown in Figure 1 and Figure 2, respectively, were each used to determine flow rate. For the diameter ratio tested in lab, The orifice plate is easiest for determining flow rate because it has a lower relative uncertainty, 0.7503% compared to the nozzle which has a relative uncertainty of 1.1%. The orifice also has a greater range of reynolds numbers, 5000 to maximum, compared to the nozzle which has a range of 10^4 to 10^7 . Typical values for discharge coefficient of orifice plate are about 0.6 and of nozzle are about 0.96

The most efficient device is nozzle because it has a higher discharge coefficient and less flow loss because of its geometry. The nozzle has a gradual change in diameter while the orifice plate has a sudden change in diameter, so the nozzle is more efficient.

The orifice plate is least expensive for fabrication and installation because it's geometry is less complex and therefore easier to manufacture than the nozzle. Additionally, the discharge coefficient is lesser than the nozzles meaning that the nozzle is more accurate and in return more expensive to fabricate.

Device conformance to professional standard

Table 1. 71mm disk team measurements summary (in)

71mm	Mean of groups measurements (in.)	Standard deviation (in)
e (inner thickness) (in.)	0.05875±0.0005	0.004112±0.0005
E (outer thickness) (in.)	0.13475±0.0005	0.0015±0.0005
Bevel Angle (deg.)	45±0.5	0.816496±0.5
d (Inner Diameter) (in.)	2.81275±0.0005	0.045139±0.0005

Table 2. 65mm disk team measurements summary (in)

65mm	Mean	Standard deviation
e (inner thickness) (in.)	0.048±0.0005	0.011343±0.0005
E (outer thickness) (in.)	0.199±0.0005	0.04±0.0005
Bevel Angle (deg.)	45±0.5	0±0.5
d (Inner Diameter) (in.)	2.5855±0.0005	0.048918±0.0005

The 65 mm disk does not meet the ASME standards. This is because the outer thickness (E) of the plate is too large. Using the equation $e \text{ (inner thickness)} < E < 0.05D$ where D (duct diameter) = 75mm from the ASME MFC-3M-2004 booklet we find that the 65mm disks E must be between 0.048in and 0.147in, and when measured out, $E = 0.199$ which exceeds the 0.147in limit. To understand these values and equations better please refer to figure 1 on TAD page 1

Because of this, the 71mm orifice plate disk was chosen to run the tests for the week, since it passed this ASME requirement, measuring out to $E = 0.13475$ which does not exceed the max thickness of 0.147in.

Ultimately Our team went with the 71mm inner diameter orifice plate for testing as it complied with all of the requirements set in the ASME MFC-3M-2004 standards. As opposed to the 65mm inner diameter orifice plate which was too thick to get accurate air flow readings.

Flow rates, uncertainties, and recommendation

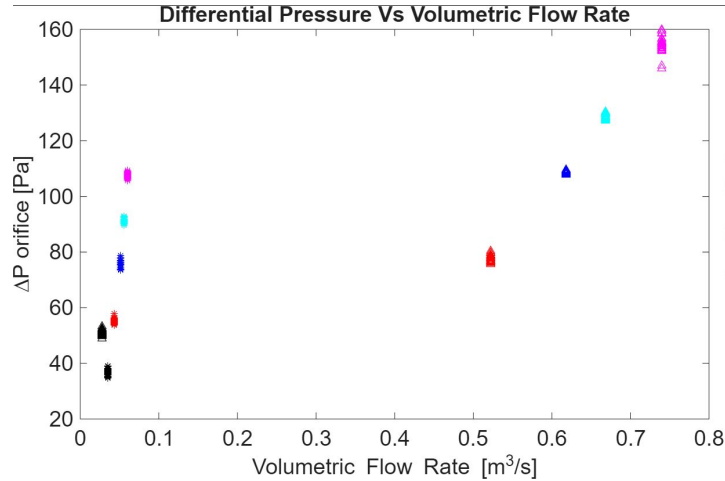


Figure 3. Differential Pressure vs Volumetric Flow Rate

Legend for Figure 3:

The orifice plate is plotted using triangles, while the nozzle is plotted using stars.

In order from trial 1 to 5 is black, red, blue, cyan, and magenta.

The instrument that should be upgraded to reduce the overall uncertainty in the flow rate is the pressure transducer. The uncertainty analysis shows that the relative uncertainty in pressure difference, $\delta(\Delta P)/\Delta P$, is much larger than the uncertainties from the other instruments, especially at the lowest flow rates where the pressure difference is small. Since volumetric flow rate is proportional to the square root of the pressure difference, the uncertainty in pressure contributes directly and dominates the total flow uncertainty. In comparison, the barometer and thermometer have very small relative uncertainties, and the caliper measurements are essentially negligible. Therefore, improving the accuracy or range of the pressure transducer would provide the greatest reduction in overall flow-rate uncertainty.

For the mNICU project, I would recommend using the orifice plate flow meter. In our testing, the orifice plate produced lower relative uncertainty at the lowest measured flow rate (about 17.7%) compared with the ISA 1932 nozzle (about 25.9%), mainly because the nozzle had a smaller ΔP at low flow, making the pressure-transducer uncertainty a larger fraction of the measurement. The orifice plate is also typically simpler and cheaper to manufacture and replace, and it is widely used with straightforward, well-established analysis procedures. While ISA 1932 nozzles can offer good performance, they generally come with higher component cost and don't provide an uncertainty advantage in our low-flow operating range unless the pressure measurement system is upgraded.

Table 3. Uncertainty measurements of the orifice plate and the nozzle at the lowest flow rate

Device (lowest flow)	V^* (m³/s)	$\delta V^*_{\max}$ (m³/s)	$\delta V^*/V^*$	Pressure Transducer $\delta(\Delta P)/\Delta P$	Barometer $\delta P/P$	Thermometer $\delta T/T$	Calipers $\delta D/D$	Calipers $\delta D_0/D_0$
Orifice Plate	0.028	0.004945	0.1766	0.3522	6.88E-05	0.001694	5.49E-06	7.69E-06
ISA 1932 Nozzle	0.035	0.009076	0.2593	0.5176	6.88E-05	0.001694	5.49E-06	7.69E-06

Sierra Manriquez, M18532809

MEEM 2901, L18, Fall

David Wallis

MAEP 1 Reflective Essay

The two assignments I chose to include were the Reverse Engineering 1 report and the System Simulation with AMESim report. The Reverse Engineering 1 Report is a group report that provides an analysis of a brad nailer, covering how the nailer is used, potentially misused, limitations, and potential design improvements. The System Simulation with AMESim report provides the findings from an AMESim spring mass damper system model simulation, where different parameter values were tested to analyze system changes. The main purpose was to prove the usefulness of simulation software in the engineering process.

The most important skills I learned are the use of simulation software to optimize the engineering process and working as a team to analyze mechanisms. I know these skills will be consistently applied in my future career. This has made it easy to continue playing with the simulation software and collaborating with my teammates to further exercise these skills. I enjoyed reverse engineering most because working with my team to disassemble the nailer and then coming up with ideas of how to improve the nailer felt natural to fulfill each other's curiosity and add on to our ideas together.

The hardest assignment this semester was the Material Properties Assessment Lab. The most challenging part was analyzing data with MATLAB. Since each tensile test was different, a new approach was needed to fit each set of data that was analyzed. To get better at analyzing data with MATLAB, I studied each MATLAB lesson from class and read through the MATLAB documentation to learn additional commands I thought would be helpful to use in my program.

Outside of MEEM 2901, the course that has helped me most is Statics. This course made the bridge truss labs my favorite because I saw my learning in a real experiment. I could calculate where the zero-force members of the truss were, then confirmed by measuring force during the experiment.

Finally, this semester, I learned that the study habits I established were a great decision. I always started my assignments the day they were assigned so that I had optimal time to complete them. This helped me put my best work into each assignment because I could maximize revision time. One thing that I want to change about my studying habits is regularly going to office hours to get questions answered instead of writing an email. I found that working face-to-face with others helps me learn better, so I believe that meeting with my professors for any class when I am welcome will help me with my conversation and collaboration skills while also getting my questions answered.



TO : David Wallis
FROM: Addie Behling, Johnathan Kengel, Sierra Manriquez
DATE : 11/11/2025

SUBJECT: Truss Analysis- Concentrated and Distributed Load

This truss analysis report is about experimental results of testing load sensors in a bridge during an applied force and hand calculations of loads for an applied force. This report details the experimental processes and results as well as comparing those results to the calculated load values.

After analysis and comparison, the calculated values vary largely when compared to the experimental values. This is because the calculations were simplified to one side carrying the load while in the experiment, the force was distributed across two sides.

This report concludes that calculated values for loads within a system do not provide accurate predictions for the real loads within the system because of the oversimplification of the three-dimensional bridge down to two dimensions. Calculated values can only provide general estimations due to the various changes around the system when in the lab and displacement of the system.

MEEM2901

Mechanical Engineering Practice Course – I

Practice Session Report- Truss Analysis- Concentrated and Distributed Load

Date: 11/11/2025

Lab Section: Thursday, 3:00

Student name: Addie Behling

Johnathan Kengel

Sierra Manriquez

Introduction

The truss analysis lab is about building a truss bridge and using load sensors to get the internal forces of the bridge. This lab serves multiple purposes, the first one being to analyze the design of a truss and how it supports weight. The second is to practice hand calculating forces using statics to compare experimental data. The lab is about analyzing theoretical and measured data, determining differences, and finding an explanation for those differentiations.

Methodology

When setting up the lab, a small-scale bridge was constructed. The bridge itself is about 68 cm long, 17 cm tall, and 12 cm wide, with small feet that stick out 5.5 cm from the bottom corners. The bridge is composed of plastic beams, half round connectors, and metal screws. The bridge design is a simple truss, having four 17 cm beams along the bottom and two along the top with three 17 cm beams perpendicular to the top and bottom. The design also has four 24 cm beams that are angled; two are on the ends angled inward, connected to the bottom and angles 45 degrees up to connect to the top. The other two 24 cm beams mimic the same but moved inward. This two-dimensional design is duplicated and connected on the other side. The two halves are connected by five 11.5 cm perpendicular beams along the five bottom joints and three additional beams along the three top joints. The design also includes four 5.5 cm beams to protrude downward from the bottom four corners and act as a stand for the bridge [1].

To calculate the correction factors for the four load sensors, three tests were conducted for each cell. The first test was measuring a weight of about 5 lb using a calibrated sensor and then measuring using the load cell. The second and third tests were identical to the first except the weight increased to 10lb in the second test and then 15 lb in the third test. After the tests were completed, the correction factor was computed for each of the three tests using Equation 1, where actual is the load measured with the calibrated sensor and measured is the load measured on the load sensor [2].

$$\text{correction factor} = \frac{\text{actual}}{\text{measured}} \quad (1)$$

The average correction factor was computed for each sensor using the three individual correction factors values for each sensor.

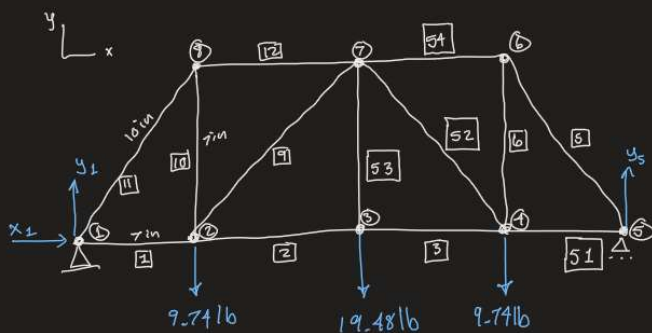
Table 1. Measured Load and Corresponding Correction Factor for Each Load Cell

Load Sensor	Measured Weight of the Loading Bags			Load Values Measured with the Load Sensors			Average Correction Factor
	Test 1	Test 2	Test 3	Test 1	Test 2	Test 3	

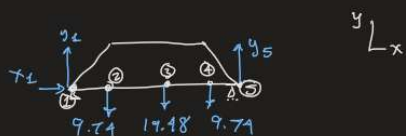
51	4.81	9.74	14.55	5.3	10.7	15.9	0.911
52	4.81	9.74	14.55	5.39	11.15	16.6	0.8808
53	4.81	9.74	14.55	5.75	11.24	16.7	0.8581
54	4.81	9.74	14.55	5.52	11.4	16.89	0.8662

The bridge was tested in two different ways. The first method was a concentrated test with 20 lb placed along the center of the bridge. This method tests the forces when each side holds up 10 lb and all the weight is concentrated on a single point. The second method was a distributed test with 20 lb in the middle and 10 lb on each side of the bridge. This method tests the forces in the bridge when each side holds up 20 lb and the weight is distributed along three points of the bridge. For both tests, the forces within the bridge and the displacement were measured. The load sensors measured the force within the bridge while a dial gauge was used to measure the displacement. Dial gauges work by using a pin that registers small linear movement and translates that movement to a dial which shows the displacement.

Hand Calculations



whole truss:



$$\sum F_x = 0 \Rightarrow x_1 = 0$$

$$\sum F_y = 0 \Rightarrow y_1 + y_5 - 9.74 - 9.74 - 19.48 = 0$$

$$\sum M_1 = 0 \Rightarrow -[7 \times 9.74] - [14 \times 19.48] - [21 \times 9.74] + [y_5 \times 28] = 0$$

$$-545.44 + 28y_5 = 0$$

$$28y_5 = 545.44$$

$$y_5 = 19.48 \text{ lb}$$

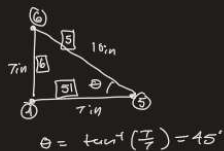
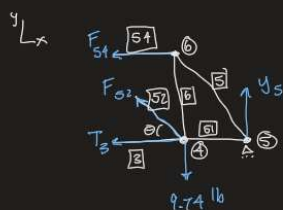
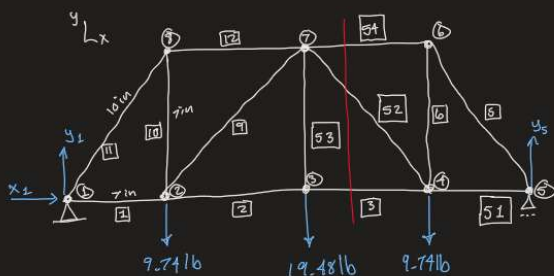
$$\rightarrow y_1 + y_5 - 9.74 - 9.74 - 19.48 = 0$$

$$y_1 + 19.48 - 38.96 = 0$$

$$y_1 - 19.48 = 0$$

$$y_1 = 19.48$$

Method of sections:



$$\sum F_y = 0 \Rightarrow F_{52} \sin \theta - 9.74 + Y_5 = 0$$

$$F_{52} \sin \theta = 9.74 - Y_5$$

$$F_{52} = \frac{9.74 - Y_5}{\sin \theta}$$

$$F_{52} = \frac{9.74 - 19.48}{\sin(45)} = -13.7744 = F_{52}$$

$$\sum M_5 = 0 \Rightarrow [F_{51} * 7] - [F_{52} \sin \theta * 7] + [9.74 * 7] = 0$$

$$7F_{51} = (7 \sin \theta F_{52}) - (9.74 * 7)$$

$$F_{51} = \frac{(7 \sin(45) * -13.7744) - (9.74 * 7)}{7}$$

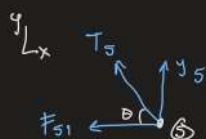
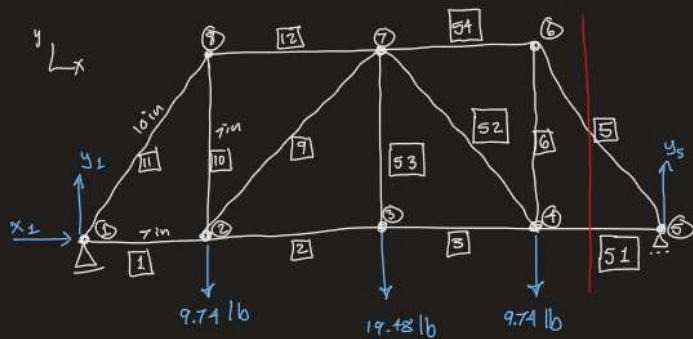
$$F_{51} = -19.48$$

$$\sum F_x = 0 \Rightarrow -T_3 - F_{51} - F_{52} \cos \theta = 0$$

$$T_3 = -F_{51} - F_{52} \cos \theta$$

$$T_3 = -(-19.48) - (-13.7744) \cos(45)$$

$$T_3 = 29.2200$$



$$\sum F_y = 0 \Rightarrow T_5 \sin \theta + y_5 = 0$$

$$T_5 = \frac{-y_5}{\sin \theta}$$

$$T_5 = \frac{-19.48}{\sin(45)}$$

$$T_5 = -27.5189$$

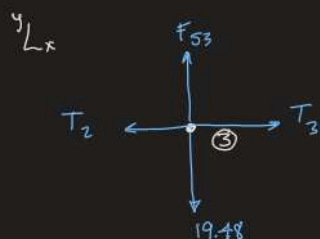
$$\sum F_x = 0 \Rightarrow -F_{51} - T_5 \cos \theta = 0$$

$$F_{51} = -T_5 \cos \theta$$

$$F_{51} = -(-27.5189) \cos \theta$$

$$F_{51} = 19.48$$

Joint 3:

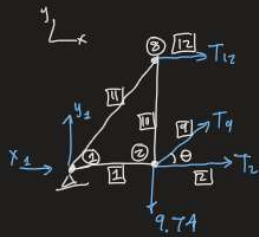
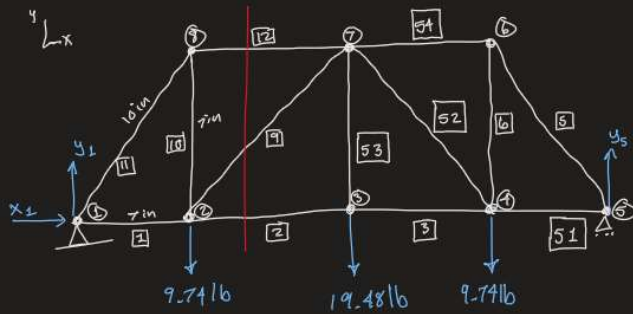


$$\sum F_y = 0 \Rightarrow F_{53} - 19.48 = 0$$

$$F_{53} = 19.48$$

$$\sum F_x = 0 \Rightarrow T_2 - T_3 = 0$$

$$T_3 = T_2 = 29.2200$$



$$\sum F_y = 0 \Rightarrow y_1 - 9.74 + T_9 \sin \theta = 0$$

$$T_q \sin \theta = 9.74 - y_1$$

$$T_9 = \frac{9.74 - 19.48}{\sin(45)}$$

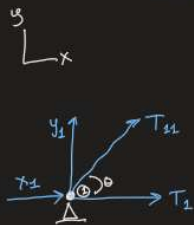
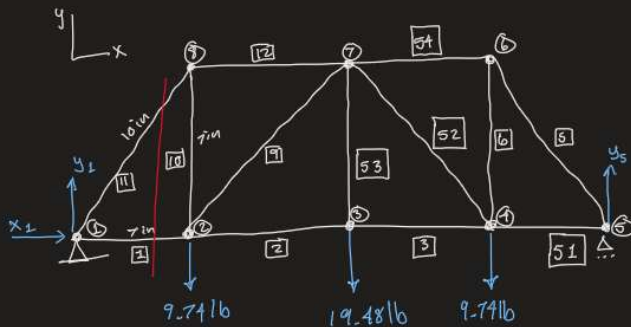
$$T_9 = -13.7744$$

$$\Sigma F_x = 0 \Rightarrow x_1 + T_2 + T_9 \cos \theta + T_{12} = 0$$

$$0 + 29.2200 + T_9 + T_{12}$$

$$T_{12} = -29.2200 - (-13.7744)$$

$$T_{12} = -15.4456$$



$$\sum F_y = 0 \Rightarrow y_1 + T_{11} \sin \theta = 0$$

$$T_{10} = \frac{-y_1}{\sin \theta}$$

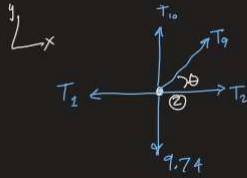
$$T_{11} = -27.5489$$

$$\Sigma F_x = 0 \Rightarrow T_1 + T_2 + T_{11} \cos \theta = 0$$

$$\frac{0 + 27.5489}{\cos(45)} = T_1$$

$$\boxed{T_1 = 38.96}$$

Joint 2 :



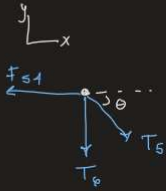
$$\sum F_y = 0 \Rightarrow T_{10} + T_9 \sin \theta - 9.74 = 0$$

$$T_{10} = -T_9 \sin \theta - 9.74$$

$$T_{10} = -(-13.7744) \sin(45) + 9.74$$

$$T_{10} = 19.48$$

Joint 6:



$$\sum F_y = 0 \Rightarrow -T_6 - T_5 \sin \theta = 0$$

$$T_6 = -T_5 \sin \theta$$

$$T_6 = -13.7744$$

Final Answers:

$$T_1 = 38.96$$

$$T_2 = 29.2200$$

$$T_3 = 29.2200$$

$$T_{51} = 19.48$$

$$T_5 = -27.5489$$

$$T_6 = -13.7744$$

$$F_{52} = -13.7744$$

$$F_{53} = 19.48$$

$$T_9 = -13.7744$$

$$T_{10} = 19.48$$

$$T_{11} = -27.5489$$

$$T_{12} = -15.4456$$

$$F_{54} = -19.48$$

Results

During this experiment, the displacement for a concentrated load was found to be a lot greater than the displacement of a distributed load. This can be shown in figures 2 and 4, comparing the gauge after both a concentrated load and distributed load was applied to the bridge. Figure 1 shows the initial gauge reading before any weight is applied and reads about .038 inches. In Figure 2, the gauge reads about 0.093 inches. This indicates that the bridges' displacement changed by 0.055 inches due to a concentrated load. The difference between readings in figures 3 and 4 is about 0.005, which indicates that the trusses displacement barely changed in result of a distributed load. Finally, this difference can be shown in the graphs shown in figures 5 and 6, where the sensors measured different force values based on where the load was placed.



Figure 1. Gauge Reading Before Applying a Concentrated Load



Figure 2. Gauge Reading After Applying Concentrated Load



Figure 3. Gauge Reading Before Applying Distributed Load



Figure 4. Gauge Reading After Applying Distributed Load

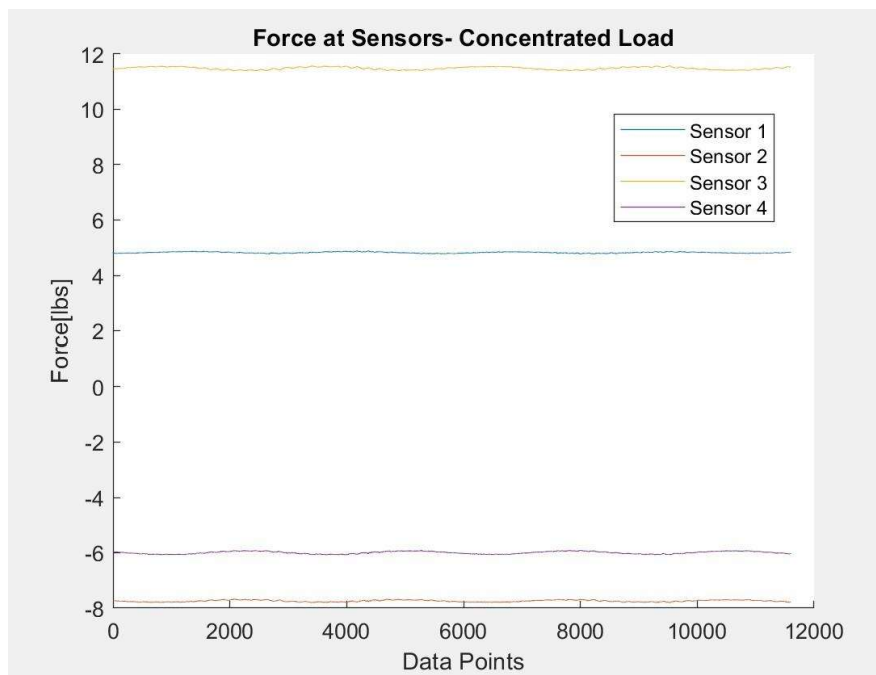


Figure 5. Graph of sensor readings for a concentrated load.

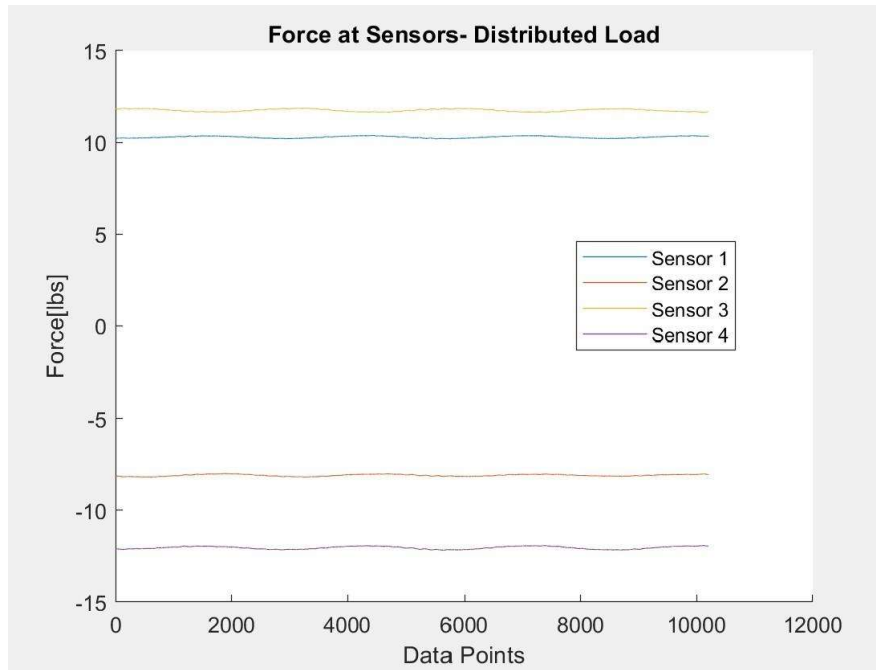


Figure 6. Graph of sensor readings for a distributed load.

Table 2. Comparison of the experimental and analytical solutions for a distributed load

Sensor #	Experimental	Analytical	Percent Error
1	10.35	19.48	46.87%
2	-8.05	-13.7744	41.56%
3	11.67	19.48	40.09%
4	-12.07	-19.48	38.04%

Discussion

The discrepancies found in Table 2 could be a result of a variety of reasons. The main source of error would be that the analytical data values for the load in each member was calculated using a two-dimensional sketch of the bridge while, in the experiment, the bridge is three-dimensional with the force distributed across two identical sides. This is shown by the percent error being around 50% for each value, meaning that if the two-dimensional side of the bridge were doubled, the values for the forces would be cut in half, causing those analytical values to match their corresponding experimental values closely.

Another factor that could have caused significant error was the setting up of the experiment. For example, the weights that were applied were not exactly 10 and 20 pounds but varied and tended to be less than what was used in the calculations. There could have also been mistakes with calculating and using the calibration correction factor and the measurement precision. Other reasons for discrepancies could include the environment that was used for testing is not equivalent to what was assumed during calculations and the potential differences between the physical model and the analytical solution model.

Conclusion

Overall, the analytical solution did not accurately present the bridge's behavior when experimentally loaded. Although it could be used as a good estimate, the calculations do not consider uneven loads and environmental factors. The calculations also do not account for the second side of the three dimensional bridge. To improve this study in the future using equal load amounts as compared to calculations is suggested because using more accurate and precise measurement equipment and limiting any potential errors while building the truss structure and calibrating the sensors.

References

- [1] "Practice Session – Bridge Concentrated Distributed - Instructions", in ME2901 canvas course, last accessed on 11/11/2025
- [2] "Lesson11A - Bridge1 - Handout", in ME2901 canvas course, last accessed on 11/11/2025



Interoffice Memorandum

TO : David Wallis
FROM: Sierra Manriquez
DATE : 11/20/2025

SUBJECT: Truss Simulation Analysis

The purpose of this memo is to analyze and compare the results of the bridge truss loading practice session. The analytical predictions, experimental measurements, and simulation results were compared to determine how accurately each model represents the real-world behavior.

The analysis found that while the hand calculations and simulation models predicted the general behavior of the bridge truss, the values did not match the experimental results. There were major discrepancies caused by simplifying the truss into two dimensions for the simulation and hand calculations. There were also sensor calibration errors that caused the data to skew.

The results indicated that the analytical model showed the truss behavior but was not accurate due to the model's limitations. Future improvements, such as 3D modeling and application of the correction factor, would help the simulations predict the physical model's performance more accurately.

MEEM2901

Mechanical Engineering Practice Course – I

Truss Analysis Simulation

Date: 20-11-2025

Lab Section: Thursday, 03:00

Student name: Sierra Manriquez

Introduction

This week's practice session was about testing and analyzing different loads on a bridge truss. In the lab, students applied distributed loads at key joints on the bridge truss, then analyzed the forces, stresses, and displacements within the structure to assess the performance and safety of the truss. Students then simulated the experiment in Simcenter 3D and MATLAB to validate the experimental results and find differences.

FEA Model Development

In MATLAB, the bridge truss was modeled as a two-dimensional object where the cross-sectional area, modulus of elasticity, nodal coordinates, and applied forces were considered to calculate the displacement of the members. The cross-sectional area and modulus of elasticity are defined at the beginning of the program, where the cross-sectional area is of the members used to build the truss during the experiment, and the modulus of elasticity was found during the tensile test practice session. The cross-sectional area is defined in units of inches squared, and the modulus of elasticity is defined in units of psi. Next in the program, the nodal coordinates are defined in pairs, where the distances between each node are measured in inches. Lastly, the applied forces were defined for the node that they would be acting on.

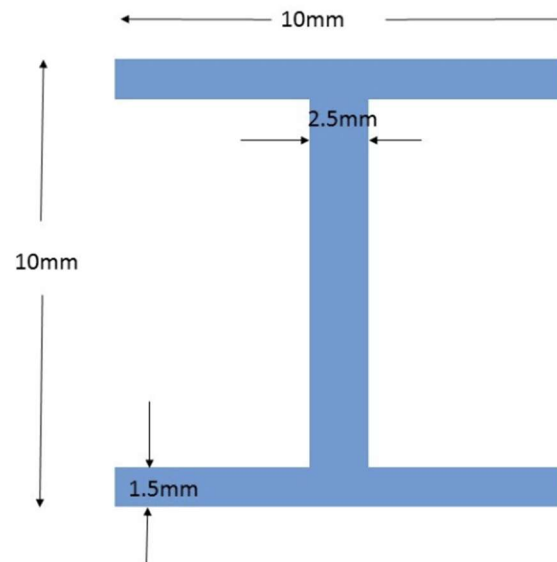
The cross-sectional area of a bridge member that would fail when the bridge truss is tested for the distributed load was calculated using equation 1, where the cross-sectional area is equal to the experimental force found during the bridge truss loading practice session over the rupture stress.

$$A_{cross-sect} = \frac{F_{experimental}}{\sigma_{rupture}} \quad (1)$$

The rupture stress used in the calculation is the ultimate stress for ABS plastic as determined in the tensile test practice session. The rupture stress is equal to 4150 psi. The experimental force used in the calculation is 11.2927 lb.

The cross-sectional area that was used in the concentrated and distributed loads practice session was calculated from the bridge members given in the lab, which all had the same cross-sectional area. Figure 1 shows the dimensioned drawing of the cross-sectional area of the members.

Figure 1. Bridge Member Cross-section Drawing [1]



To calculate the cross-sectional area of the members, the method of composite shapes was utilized. The area was split into three rectangles, where the area of each rectangle was calculated and then summed to give the total area of the cross-section. The calculated cross-sectional area is 0.073 inches squared.

Table 1. Material Properties of the Truss

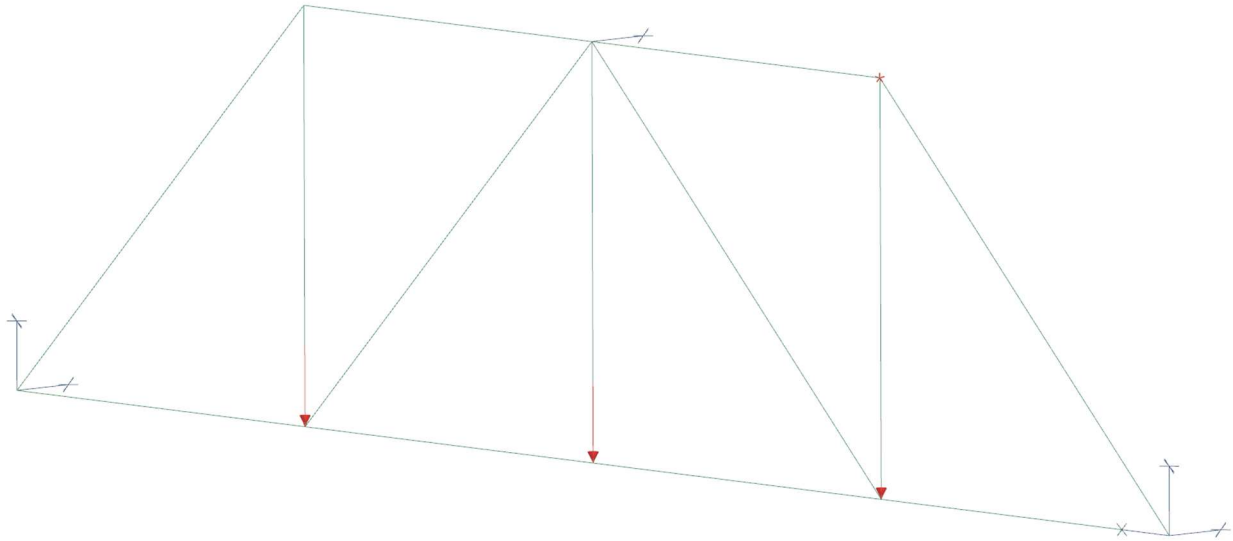
Material	Young's Modulus	Poisson's Ratio	Yield Strength	Ultimate Strength
ABS	20000000 kPa	0.35	40000 kPa	4150 psi

The geometric parameters of the bridge truss as a whole are 28 inches in length, 4.5 inches in width, and 7 inches in height. Each Individual member has a cross-sectional area of about 0.229 inches squared [1].

The two load distributions that were tested in the bridge truss loading practice session were simulated in Simcenter 3D. The first test was a 20 lb. concentrated load located on the bottom center of the joint of the truss. The second test was a 40 lb. distributed load where 20 lb. is located on the bottom center joint, and two 10 lb. loads are located on the joints on either side of the center.

The Simcenter 3D model contains multiple assumptions, boundary conditions, and constraints. The simulation assumes ideal conditions where the loads are exact, and each member and joint of the truss behaves ideally. In the simulation, each member is idealized into a two-dimensional rod element. The mesh of elements is constrained at the connections with the environment and at the top, center node. Figure 2 shows the mesh and constraints of the Simcenter 3D model as well as the three forces for the distributed load test.

Figure 2. Geometric FEA Truss Model Mesh and Constraints

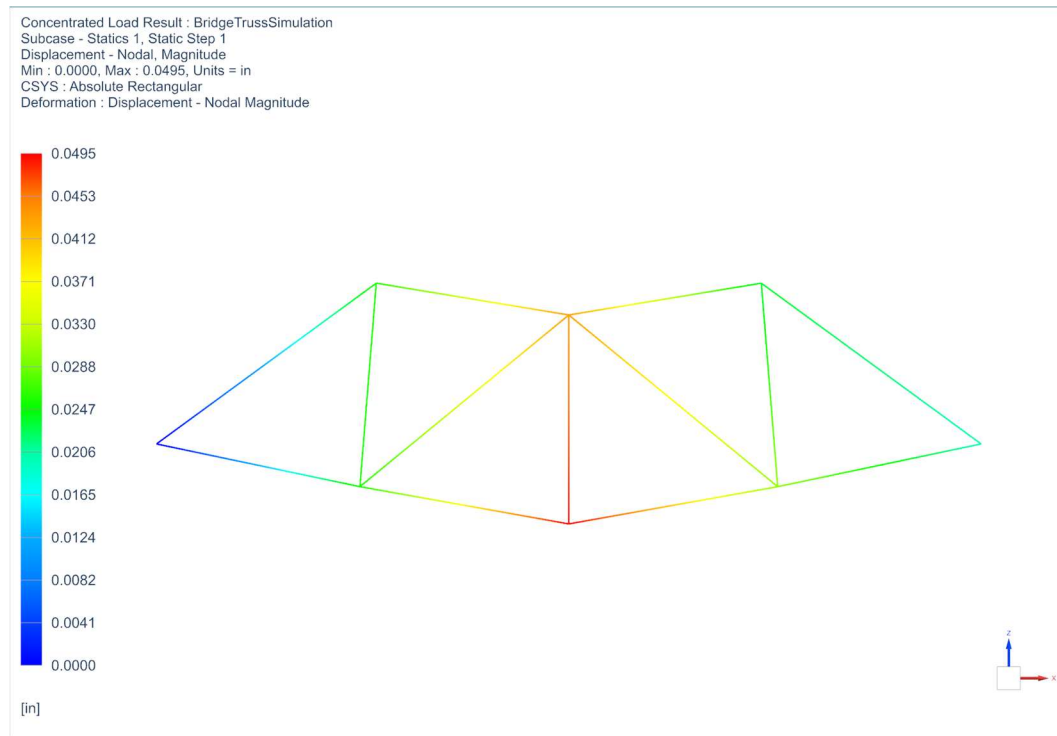


Results

Simcenter 3D Simulation Results

When the tests were simulated in Simcenter 3D, a plot of displacement of the truss was generated to show how the bridge truss would move when loaded. Figure 3 shows the simulated plot of displacement of the bridge for the first test, where a 20 lb concentrated load was located on the bottom center joint of the truss.

Figure 3. Displacement Plot of Concentrated Load

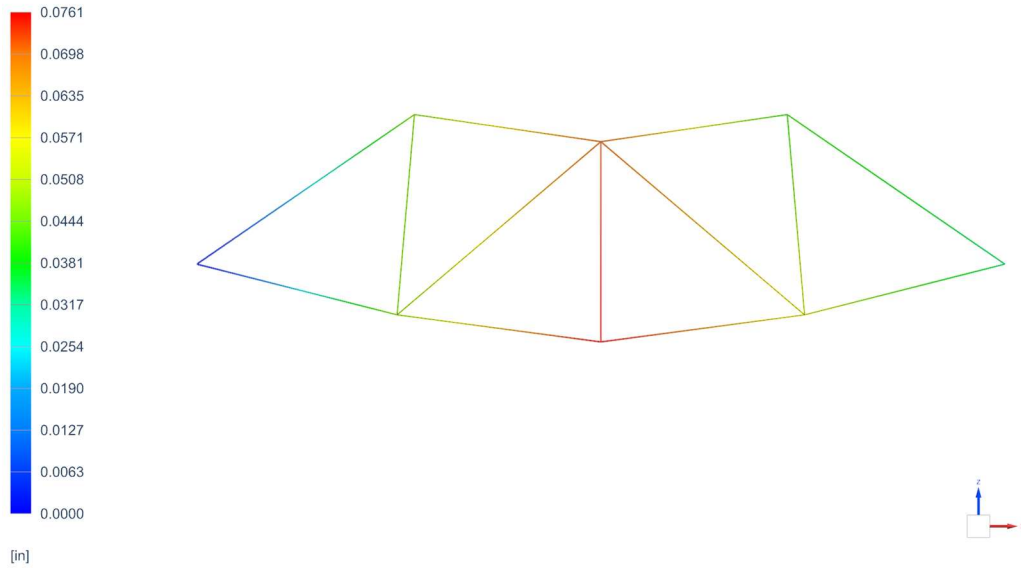


The plot of the concentrated load shows the displacement of the members in the bridge truss, where the blue coloration represents small displacement and the red coloration represents large displacement. The displacement is measured in units of inches. For the 20 lb concentrated load, the displacement is greatest at the center of the bridge truss, where the load is applied, and least at the connections to the environment. The displacement is measured in units of inches.

Figure 4 shows the simulated plot of displacement of the bridge truss for the second test where a distributed load of 40 lb was applied.

Figure 4. Displacement Plot of Distributed Load

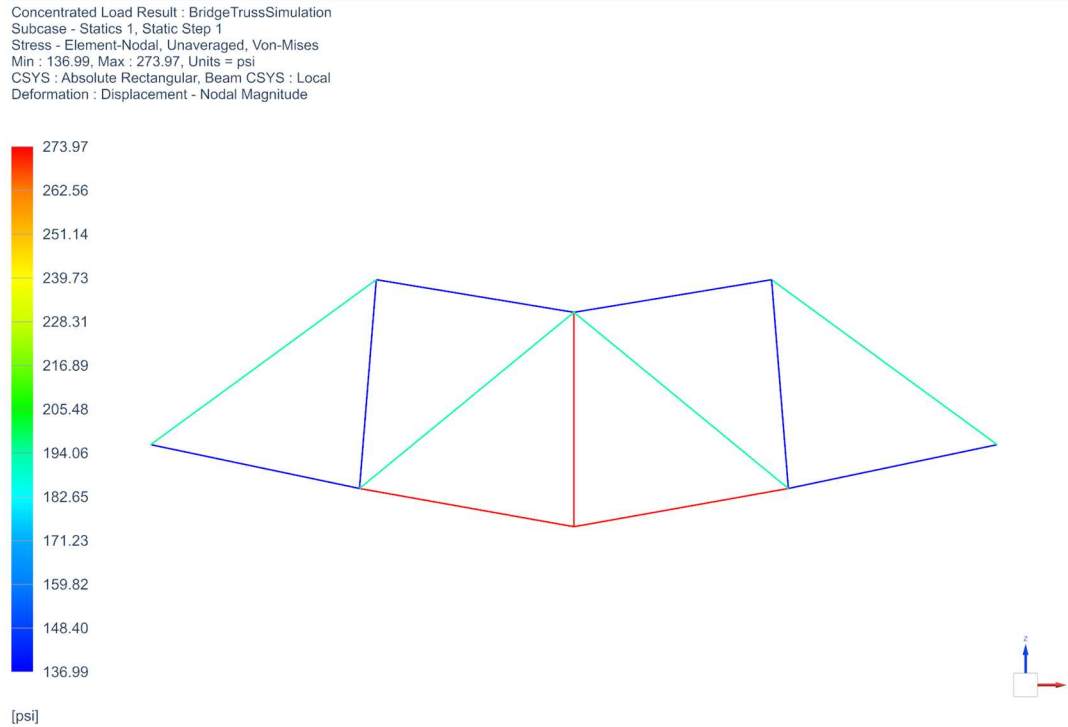
Distributed Load Result : BridgeTrussSimulation
Subcase - Statics 1, Static Step 1
Displacement - Nodal, Magnitude
Min : 0.0000, Max : 0.0761, Units = in
CSYS : Absolute Rectangular
Deformation : Displacement - Nodal Magnitude



The distributed load plot is similar to the concentrated load, where the greatest displacement is located at the center of the truss, where the greatest load is applied, and the least displacement is located at the connections with the environment. The displacement color key located on the left-hand side of the figure shows the numerical displacement value associated with each color. The displacement is measured in units of inches.

The plot of stress of the truss members for the concentrated and distributed load was also generated in Simcenter 3D. Figure 5. Shows the plot of stress of the truss members for the concentrated load and Figure 6. shows the stress of the truss members for the distributed load.

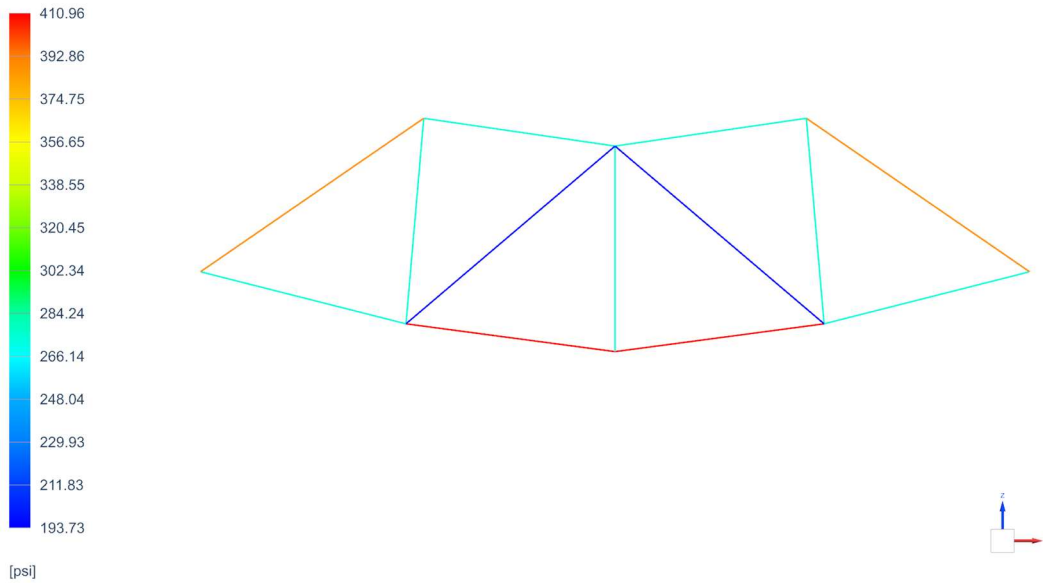
Figure 5. Stress Plot of Concentrated Load



The concentrated load plot of stress is measured in units of psi, and the stress in each member is color-coded according to the scale on the left-hand side of the figure. The maximum stress occurs in the members that are connected to the joint where the concentrated load was applied. The smallest stress was located on the zero-force members and on the members that do not carry great force. The stress applied to each member was distributed evenly throughout the member.

Figure 6. Stress Plot of Distributed Load

Distributed Load Result : BridgeTrussSimulation
 Subcase : Statics 1, Static Step 1
 Stress : Element-Nodal, Unaveraged, Von-Mises
 Min : 193.73, Max : 410.96, Units = psi
 CSYS : Absolute Rectangular, Beam CSYS : Local
 Deformation : Displacement - Nodal Magnitude

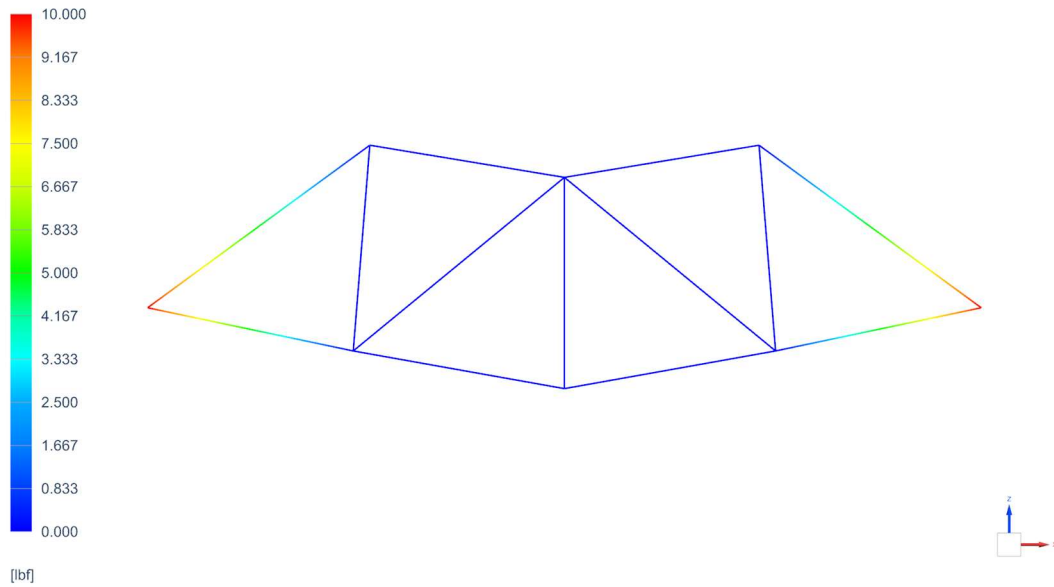


The plot of the stress of the truss members for the distributed load is very similar to the concentrated load stress plot. The maximum stress is located where the largest load was applied, and the least stress is located at the members that carry the smallest amount of force. The stress of the truss members is measured in units of psi.

A diagram of the reaction forces of the bridge truss was generated in Simcenter 3D for both the concentrated and distributed loads. Figure 7 shows the reaction force diagram for the concentrated load, and Figure 8 shows the reaction force diagram for the distributed load.

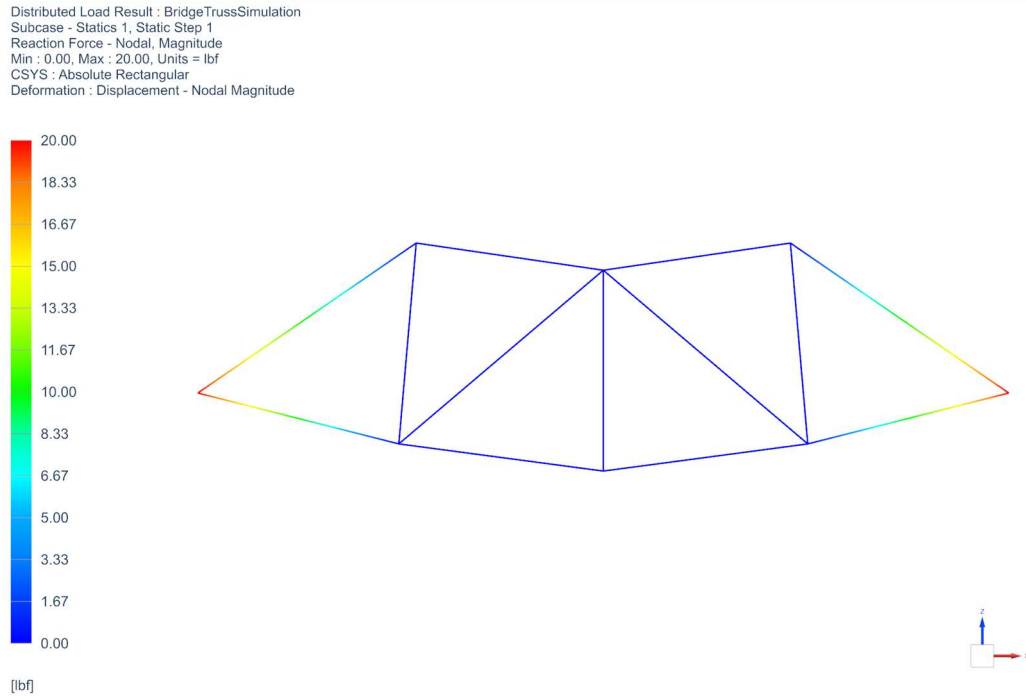
Figure 7. Reaction Force Diagram of Concentrated Load

Concentrated Load Result : BridgeTrussSimulation
 Subcase : Statics 1, Static Step 1
 Reaction Force - Nodal, Magnitude
 Min : 0.000, Max : 10.000, Units = lbf
 CSYS : Absolute Rectangular
 Deformation : Displacement - Nodal Magnitude



The reaction force diagram of the concentrated load shows where the reaction forces with the environment are distributed throughout the bridge truss. The reaction forces are measured in units of lbf. The reaction forces are at maximum where the bridge truss connects to the environment, and the reaction forces are at minimum on the members that do not have reactions with the environment.

Figure 8. Reaction Force Diagram of Distributed Load

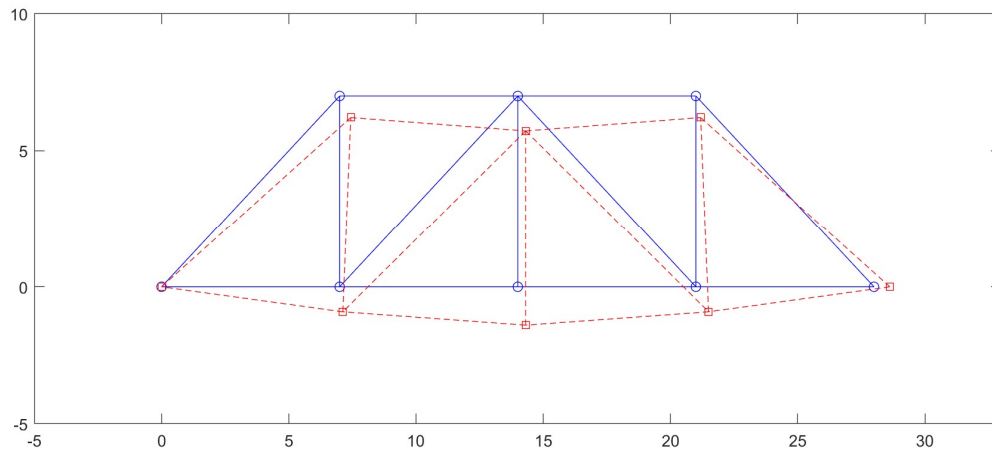


The distributed load diagram of the reaction forces is the same as the concentrated load diagram, where the maximum reaction force is located nearest to the point where the reaction forces with the environment are located, and the minimum reaction force is located furthest from the point where the reaction forces are located. The reaction forces are measured in units of lbf.

MATLAB Analysis

The distributed load was modeled in MATLAB using the values for modulus of elasticity, cross-sectional area, node coordinates, and applied force that were used in each of the other simulations and calculations. Figure 9 shows the plot of the displacement of the truss under the distributed load.

Figure 9. MATLAB Plot of Exaggerated Displacement of the Truss



The blue lines create the initial position of the bridge truss before loading occurs, and the red dotted lines create the displaced truss after loading has been applied. Since the distributed load is applied nearest to the center node, the center has the most displacement.

Comparative Analysis

The following tables show the relationship between how closely the analytical, experimental, MATLAB, and Simcenter 3D results relate to each other. The internal load and stress are evaluated for each of the thirteen members of the bridge truss. Figure 10 shows the number that corresponds to each member of the truss.

Figure 10. Labeled Members of the Truss from Analytical Hand Calculations [2]

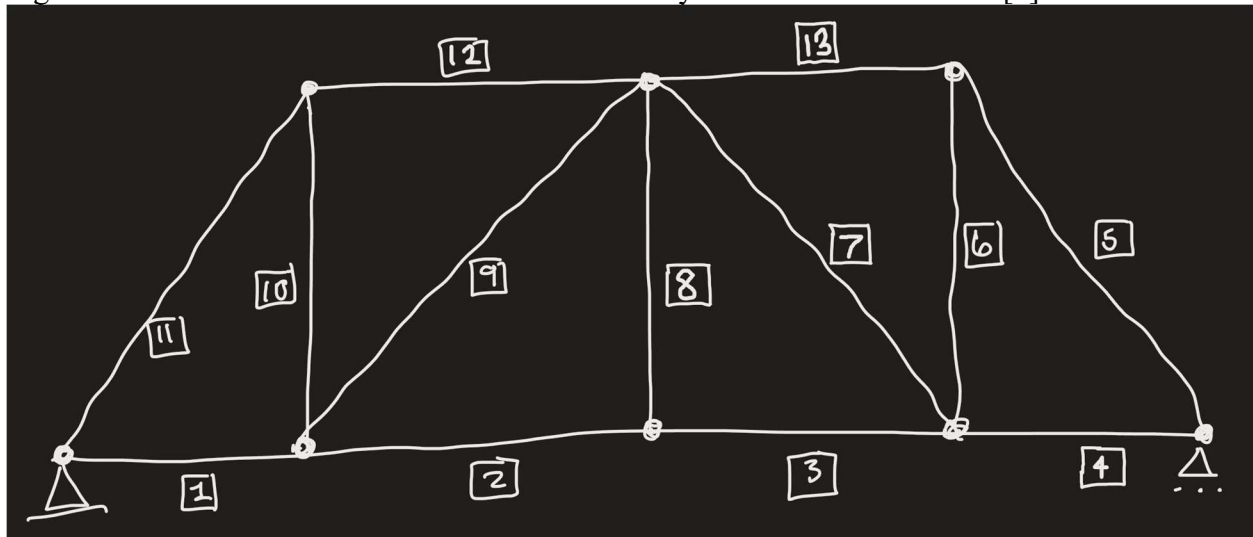


Table 2. Comparison of Internal Load in each member from Analytical, Experimental, MATLAB, and Simcenter 3D Results of Distributed Load Case

Member	Analytical	Experimental	% Error	MATLAB	% Error	Simcenter 3D	%Error
--------	------------	--------------	---------	--------	---------	--------------	--------

1	38.96	N/A	N/A	10.00	74.33	20.00	48.67
2	29.22	N/A	N/A	15.00	48.67	29.93	2.43
3	29.48	N/A	N/A	15.00	49.12	29.93	1.53
4	19.48	10.35	46.87	10.00	48.67	20.00	2.67
5	27.55	N/A	N/A	14.14	48.68	28.28	2.65
6	13.77	N/A	N/A	10.00	27.39	20.00	45.24
7	13.77	8.05	41.56	7.07	48.67	14.14	2.69
8	19.48	11.67	40.09	10.00	48.67	20.00	2.67
9	13.77	N/A	N/A	7.07	48.67	14.14	2.69
10	19.48	N/A	N/A	10.00	48.67	20.00	2.67
11	27.55	N/A	N/A	14.14	48.68	28.28	2.65
12	15.45	N/A	N/A	10.00	35.28	20.00	29.45
13	19.48	12.07	38.04	10.00	48.67	20.00	2.67

The differences in Internal Load for each calculation are quantified into a percentage error representing how far the calculated value deviates from the Analytical value. For each set of results except for the analytical results, there is a corresponding % Error column immediately to the right, which presents the quantified difference from the analytical results. The experimental values were only measured in four specific members therefore, the experimental values for the internal load do not exist for the members that did not have a sensor attached. The internal load for each member is measured in units of lbf.

Table 3. Comparison of Stress in each member from Analytical, Experimental, MATLAB, and Simcenter 3D Results of Distributed Load Case

Member	Analytical	Experimental	% Error	MATLAB	% Error	Simcenter 3D	%Error
1	533.70	N/A	N/A	136.99	48.67	273.97	48.67
2	400.27	N/A	N/A	205.48	2.67	410.96	2.67
3	400.27	N/A	N/A	205.48	2.67	410.96	2.67
4	266.85	141.78	46.87	136.99	48.66	273.97	2.67
5	377.38	N/A	N/A	193.73	48.66	387.46	2.67
6	188.69	N/A	N/A	136.99	27.40	273.97	45.20
7	188.69	110.27	41.56	96.86	48.67	193.73	2.67
8	266.85	159.86	40.09	136.99	48.66	273.97	2.67
9	188.69	N/A	N/A	96.86	48.67	193.73	2.67
10	266.85	N/A	N/A	136.99	48.66	273.97	2.67
11	377.38	N/A	N/A	193.73	48.66	387.46	2.67
12	211.58	N/A	N/A	136.99	35.25	273.97	29.49
13	266.85	165.34	38.04	136.99	48.66	273.97	2.67

The differences in Stress for each calculation are quantified into a percentage error representing how far the calculated value deviates from the Analytical value. For each set of results except for the analytical results, there is a corresponding % Error column immediately to the right, which presents the quantified difference from the analytical results. The stress in each member of the truss is measured in units of psi.

Discussion

During the experiment, there were places where the data could be skewed, causing discrepancies in the calculations as well as the simulations. A major source of error in the experimental setup is the load sensor calibrations. The load sensor calibrations are important because it allows for the correction factor to be calculated and applied to the measured loads. If the correction factor is not applied to the measured loads, then when the loads are used for calculations, all the numbers will be skewed, causing deviation from the theoretical values in the results.

An additional major discrepancy from the experimental model and the simulation models is that the simulation models are idealized into a two-dimensional truss which only accounts for the one side of the bridge truss. In the experiment, the bridge truss is three-dimensional which means that the applied load is distributed throughout two sides of the truss instead of just one. This cause for the calculations to have an error of about 50% since the experimental load is distributed across two sides.

There are many small discrepancies as well as some major discrepancies in the results. These discrepancies are likely because of the idealizations that are made for the simulations and human error while calculating and inputting the values into the simulations. The simulations in MATLAB and Simcenter 3D assume ideal loading and constraints are applied to the truss, while during the experiment, the force applied on the truss from the loading was not exact, and the truss was assembled by hand. In the hand calculations and in the software simulations, the members are idealized as two-dimensional rods, while in the experiment, the members are I-beams. This difference in shape impacts the accuracy of the results because different shapes distribute stress and strain differently.

Conclusion

This practice session let students compare and analyze the analytical, experimental, MATLAB, and Simcenter 3D results for the bridge truss loading experiment. It was found that the analytical solution followed the general behavior of the truss but did not accurately match the experimental values. The largest discrepancies in the calculated values were a result of modeling the truss as a two-dimensional system when the physical truss has load distributed across both sides in three dimensions. Other errors that caused discrepancies were a result of load sensor calibration errors, truss assembly, and differences in the shape of the cross-sectional area of the members.

To improve future results, the simulation could be an idealized three-dimensional truss to match the experiment more closely. The applied loads that were plugged into the simulation should also reflect the loads that were measured during the practice session so that the small differences in load do not skew the calculations.

References

- [1] “Practice session – bridge concentrated distributed – instructions”, in ME2901 canvas course, last accessed on 11/20/2025
- [2] S. Manriquez, A. Behling, J. Kengel, “Material Property Assessment”, unpublished



Interoffice Memorandum

TO : David Wallis
FROM: Addie Behling, Johnathan Kengel, Sierra Manriquez
DATE : 12/02/2025

SUBJECT: Truss Analysis

The failure truss analysis is about testing material structures and failure conditions as well as the use of simulated data and comparison of simulated data with experimental data. This lab report gives a thorough analysis of the different values obtained and possible reasons behind the discrepancy.

After analysis of the results, clear discrepancies were noticed between the methods while some methods were very similar to each other.

Results show that experimental data still varies from other methods of data collection. The reasons behind the differences are believed to be the result of limitations and set up of the 3D printer and the different setups of the simulation.

MEEM2901

Mechanical Engineering Practice Course – I

Truss Failure Test Analysis

Date: 02-12-2025

Lab Section: Thursday, 03:00

Student name: Addie Behling

Johnathan Kengel

Sierra Manriquez

Introduction

The truss bridge is an effective design for supporting and converting loads into forces of compression and tension, which is why trusses are often used when building bridges. Failure analysis looks for where problems can occur and determines the causes and effects of the problem. Through this analysis, one can solve any potential problems before they arise. Finite element analysis (FEA) shows its use by using a model without needing to use materials. Through the use of computer simulations, a model can be tested and adjusted without the need to create prototypes. This way, the model can be modified and optimized so money can be saved from building many simulated prototypes instead of using materials to build many physical prototypes. Then, when the model has been optimized through simulation, a prototype can be built and tested. Experimental validation is important to ensure that the simulated model is accurate and that the prototype works properly. Analyzing the differences between the experimental and simulated results helps to show possible errors in the creation of the prototype as well as potential errors in the simulation settings.

3D Printed Truss Design and Manufacturing

The bridge truss was designed as a Howe style bridge consisting of two horizontal top and bottom beams connected in between by diagonal and vertical members. One member of the truss was chosen to be redesigned to break exactly at a distributed load of 40 lb. The cross-sectional area of the designed member was calculated by hand for CAD modeling the new member and then 3D printing the member for testing. This values was achieved by using Equation 1, where the cross-sectional area is equal to the experimental force found during the bridge truss loading practice session over the rupture stress, we calculated a cross-sectional area of 0.002704 square inches.

$$A_{cross-section} = \frac{F_{experimental}}{\sigma_{rupture}} \quad (1)$$

The truss structure was made using members of ABS plastic. Because of this we had to consider material properties such as ultimate strength and the factor of safety.

Because the Factor of Safety is equal to one, the applied stress is equal to the ultimate strength, which was determined in the tensile test practice session to be 4150 psi for ABS. The experimental force in the equation is the measured force during the distributed loading case found in the first truss analysis lab, which we measured to be 10.35 pounds. We then used the calculated correction factors to incorporate the calibration errors into the sensors which gave a corrected measured force of 11.2927 lb. To create the 3D model of a truss member with a cross-sectional area of 0.002704 inches, using the Truss4-3D Print Template from canvas and connected the two member ends with a piece with dimensions 0.052 in by 0.052 in to match the calculated area.

Bridge Testing Procedure

The experimental setup was to use the same bridge design used in the truss analysis lab, positioning it across an open space. The bridge's physical design is Howe style with four horizontal members along the bottom and two along the top. Afterwards, 4 truss members were replaced with load sensors to measure the forces within the system throughout the failure process. Next, the prepared 3D printed truss described above replaced truss member 1, a member commonly called an end post brace. This member is located on the bottom left side of the bridge truss.

The next process was testing the bridge; this was done by slowly adding weight to the hooks below the bridge. The process of adding weight was to follow steps to add weight gradually waiting between each step in case of failure delay.

FEA Model Development (FEA Model in Simcenter 3D)

The FEA model was developed using the material properties of ABS consistent with the truss model used in the experiment. Key parameters included a Young's modulus of approximately 290-380 ksi, yield strength equal to 4150 psi, Poisson's ratio of 0.35, and a predicted density around 70 lb/ft³ [1].

The truss geometry was created in Simcenter 3D using the same node coordinates and member connectivity as the experimental and analytical models. Each joint in the truss was defined as a node, and individual members were assigned as a rod or beam element connection between nodes. Boundary conditions were replicated in the support configuration by fixing the bottom left node to the x, y, and z axes, the bottom right node to the y and z axes, and the top middle node to the z axis.

To simulate the applied distributed loads and constraints, the load magnitude and location were identical in both simulations to the experimental model. After defining the loads and constraints, the model was structurally analyzed to determine the node displacements, reaction forces, and member stress. Figure 1 below shows the applied load and fixity of the CROD model that was simulated in Simcenter 3D.

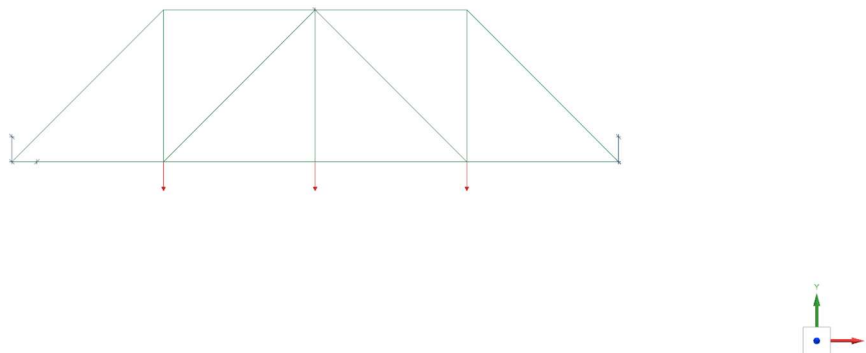


Figure 1. Screenshot of FEM model setup for CROD

Also pictured below in Figure 2 is the truss model set up that was simulated in Simcenter 3D using CBEAM elements.

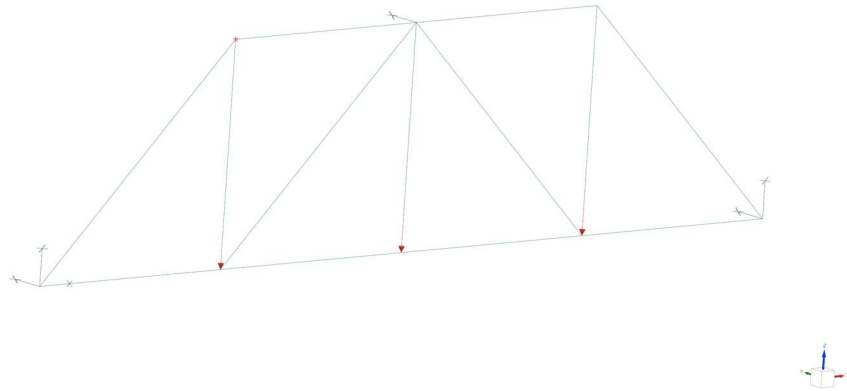


Figure 2. Screenshot of FEM model setup for CBEAM

The reaction forces in the set up for the CBEAM model are identical to the reaction forces in the CROD system set up. The bottom left side of the bridge truss is fully fixed and the bottom right side of the bridge truss is partially fixed. There is an additional partial fix in the y direction at the top center node.

Results and Discussion

Simcenter 3D Simulation Results

Figure 3 presents the plot of displacement measured in units of inches for the CROD simulation in Simcenter 3D.

Distributed Load Result : BridgeTrussSimulation
 Subcase - Statics 1, Static Step 1
 Displacement - Nodal, Magnitude
 Min : 0.0000, Max : 0.0761, Units = in
 CSYS : Absolute Rectangular
 Deformation : Displacement - Nodal Magnitude

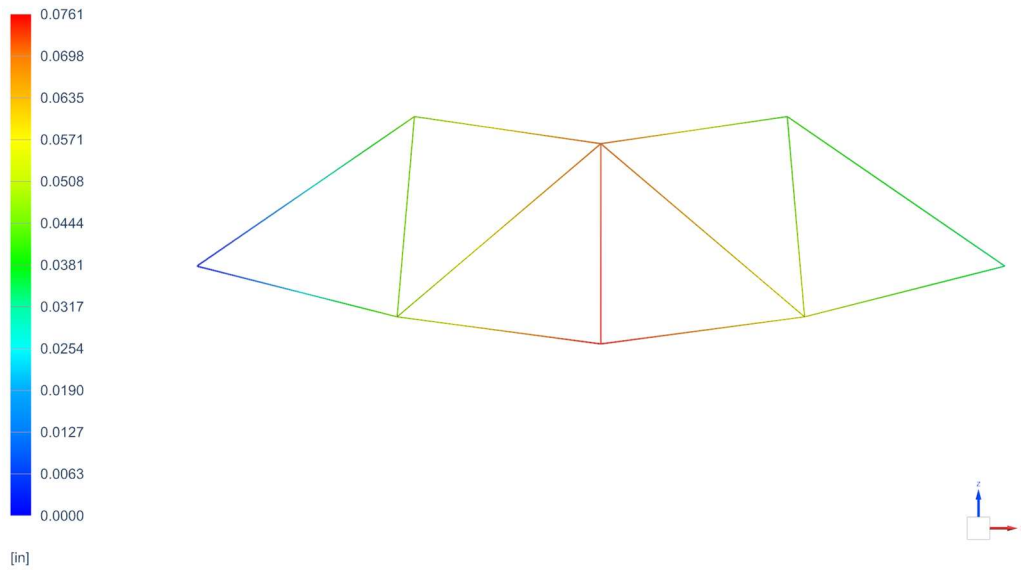


Figure 3. Plot of Displacement of Truss Using CROD

The maximum displacement is located at the center of the bridge truss, where the distributed force acts the greatest on the truss. The minimum displacement is located nearest to the connections to the environment because of the fixity of the truss preventing displacement.

Figure 4 presents the resulting plot of displacement measured in units of inches of the bridge truss using CBEAM in Simcenter 3D.

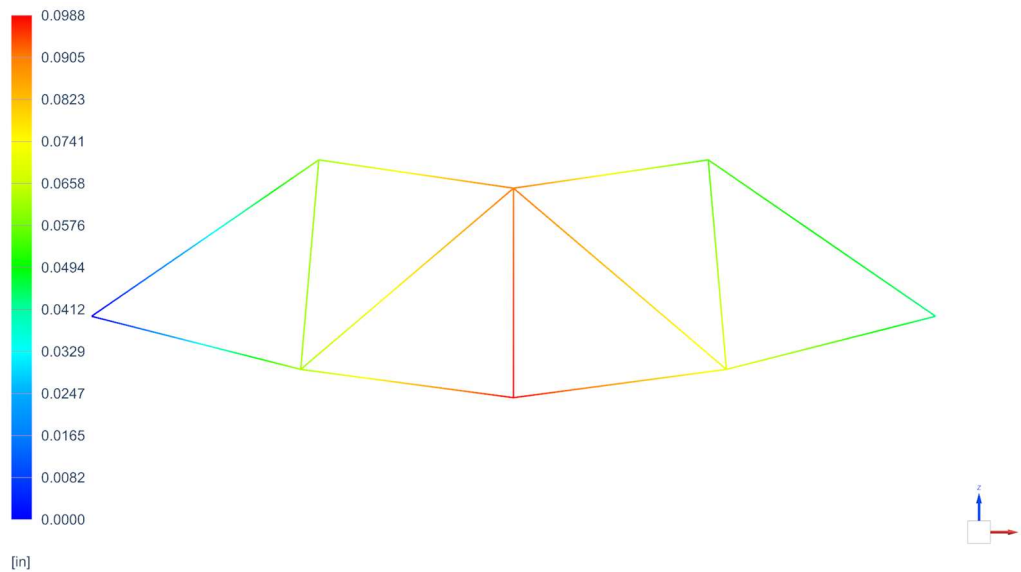


Figure 4. Plot of Displacement of Truss Using CBEAM

Like the CROD simulation, the maximum displacement is located at the center of the truss and the minimum displacement is located at the connections to the environment. The maximum displacement is slightly higher than the maximum displacement for the CROD element simulation.

Figure 5. shows the plot of the reaction forces measured in units of lbf acting on the bridge truss when simulated using the CROD elements.

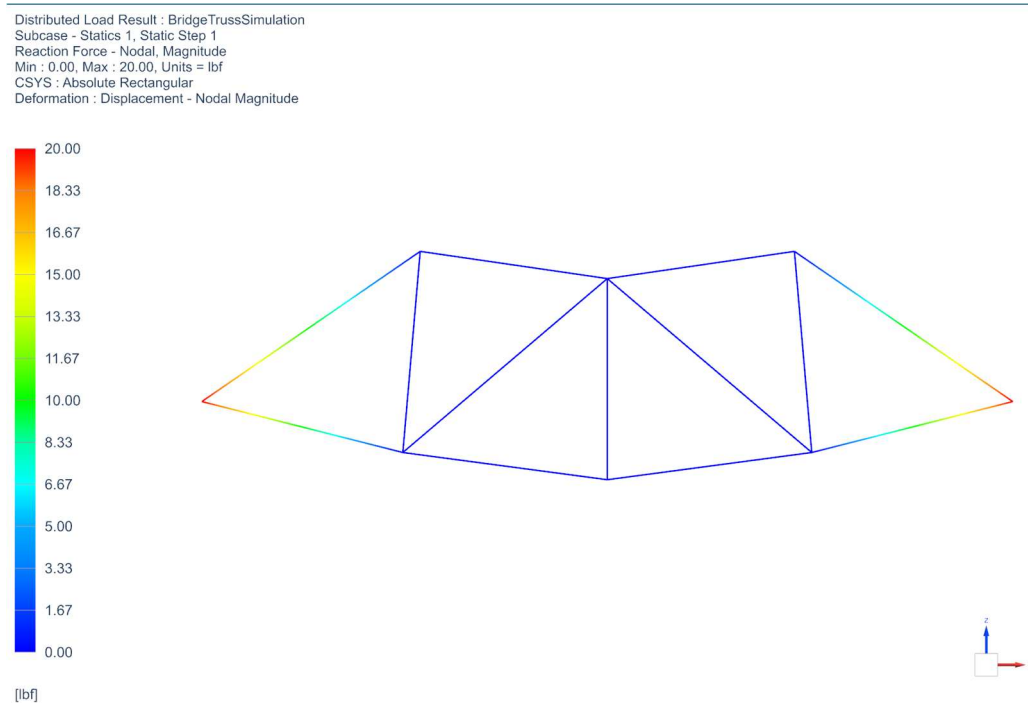


Figure 5. Reaction Force Diagram of Truss Using CROD

The greatest reaction forces acting on the truss are located nearest to the connections with the environment. This is because the reaction forces are located at the point where the truss connects to the environment.

Figure 6 shows the plot of the reaction forces measured in units of lbf of the truss using CBEAM elements.

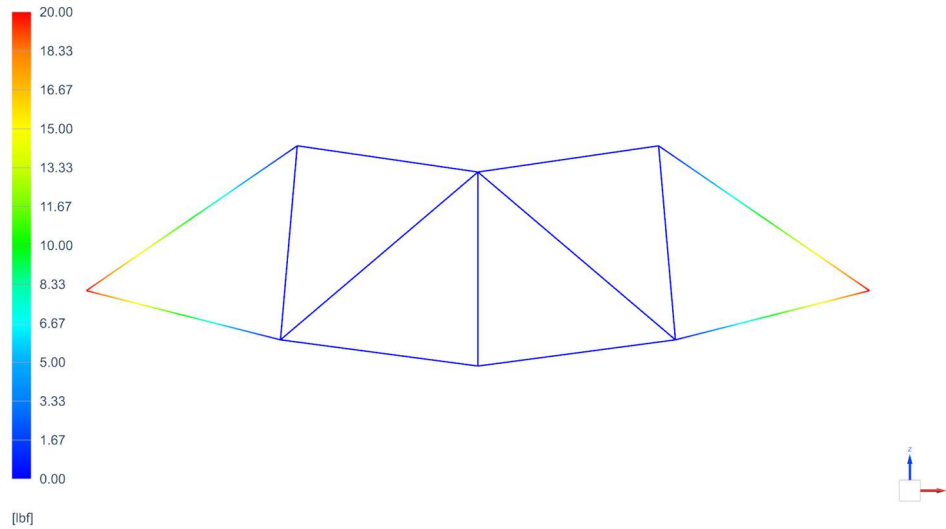


Figure 6. Reaction Force Diagram of Truss Using CBEAM

Similar to the CROD reaction force diagram, the maximum force is located at the bottom left and bottom right nodes, where the truss connects with the environment.

Experimental Results

For the first failure test, the 3D printed member broke when the load was at 20 lb. The second failure test, the 3D printed member broke sooner with a load of about 7.5 lb. Both values are less than the predicted failure load of 40 lb. This means that the 3D printed members were weaker than predicted.

Comparative Analysis

The members of the truss are identified by numbers 1-13. Figure 7. Shows the a sketch of the bridge truss and which number ID corresponds to each member.

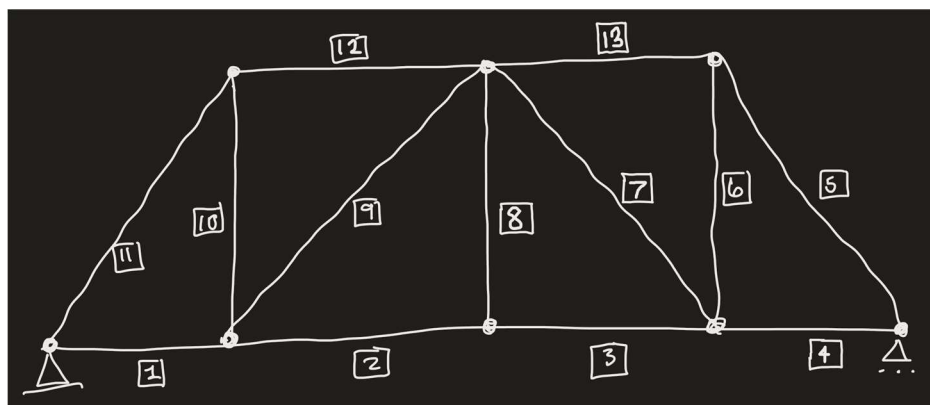


Figure 6. Reaction Force Diagram of Truss Using CBEAM

Table 1 shows each different simulation software's results for the internal loads in each member of the bridge truss.

Table 1. Comparison Between Internal Loads in Each Member

Truss Member	Analytical Results	Experimental Results	MATLAB Results	Simcenter 3D CROD Results	Simcenter 3D CBEAM Results
1	9.74	N/A	10.00	10.00	14.27
2	29.22	N/A	30.00	29.93	16.41
3	29.48	N/A	30.00	29.93	18.58
4	9.74	10.35	10.00	10.00	14.27
5	27.55	N/A	28.28	28.28	17.34
6	6.88	N/A	10.00	10.00	15.13
7	6.88	8.05	6.88	7.73	10.13
8	9.74	11.67	10.00	10.00	13.08
9	9.74	N/A	9.74	9.71	9.84
10	9.74	N/A	10.00	10.00	10.53
11	27.55	N/A	28.28	28.28	18.48
12	15.45	N/A	10.00	10.00	15.13
13	19.48	12.07	10.00	10.00	16.81

The results of each simulation and the experiment were individually compared to the analytical results to compute a percent error for each member. The percent error of each member was then averaged to get an average percent error for each simulation software. This average percent error represents how much the simulation deviates from the analytical results which are the theoretical values for each member. These results are presented in Table 2.

Table 2. Average Internal Load % Error for Each Method

	Experimental Results	MATLAB Results	Simcenter 3D CROD Results	Simcenter 3D CBEAM Results
% Error	20.28	11.31	12.45	36.17

Table 3 and Table 4 follow the same process as Table 1 and Table 2, however the measured value is now stress in each member instead of internal load.

Table 3. Comparison Between Stresses in Each Member

Truss Member	Analytical Results	Experimental Results	MATLAB Results	Simcenter 3D CROD Results	Simcenter 3D CBEAM Results
1	266.85	N/A	273.98	273.97	193.81
2	400.27	N/A	410.96	410.96	280.94
3	400.27	N/A	410.96	410.96	310.33

4	266.86	141.78	273.98	273.97	145.32
5	377.38	N/A	387.46	387.46	235.55
6	188.69	N/A	273.98	273.97	205.46
7	188.69	110.27	197.72	193.73	137.53
8	133.43	159.86	273.98	273.97	177.57
9	188.69	N/A	197.72	193.73	137.53
10	266.86	N/A	273.98	273.97	205.46
11	377.38	N/A	387.46	387.46	471.09
12	211.58	N/A	273.98	273.97	205.56
13	133.43	165.34	273.98	273.97	228.35

Table 4. Average Stress % Error for Each Method

	Experimental Results	MATLAB Results	Simcenter 3D CROD Results	Simcenter 3D CBEAM Results
% Error	33.04	24.13	23.21	8.07

Discussion

The discrepancies between the experimental results and the predicted failure loads can be a result of several factors related to the accuracy of the 3D-printed beam and the limitations of 3D printed pieces. The area of the middle cross section of the 3D printed beam is relatively small. Because it is so small, it is almost impossible for the 3D printer to print a piece to this desired level of accuracy. This lack of accuracy is a result of minor defects such as surface irregularities, voids, and incomplete filament bonding. These defects alter the actual stiffness and strength of the printed pieces. In addition, the 3D printer's resolution and dimensional tolerance create slight variations in area and geometry, which will affect the load capacity and distribution of internal forces within the truss. For us, our second member used in the experiment was hallowed due to a printing error. Because of this, the measured maximum load was much less than the calculated and simulated values, which proves that there are manufacturing and accuracy limitations of a 3D printing machine.

Discrepancies can also be found when comparing the CROD and CBEAM simulations. The rod elements are 1 dimensional, meaning that they model a limited number of forces and are good for pure tension and compression models. The beam elements are 3 dimensional, which can model bending moments, more forces, and torsion. This resulted in a higher displacement value, which better matched experimental results. Because the CBEAM simulation is able to model more similarly to how a real truss member reacts, the beam elements better approximate the actual 3D-printed geometry.

Experimental error could have also contributed to discrepancies. Errors like sensor calibration, the applied correction factor, and small variations on load amounts and increments applied can differ from calculated and simulated values. Also, physical assembly imperfections

and applied boundary conditions assumed in the hand calculations and FEA models can cause discrepancies. These differences between an idealized model and the actual experimental setup could explain why failure occurred earlier in the experiment than predicted in the simulation and hand calculations. Overall, the combination of printing errors, experimental errors, applied boundary conditions, and the differing models of CROD and CBEAM elements are a cause of the discrepancies found when comparing experimental and predicted results. These discrepancies and their causes demonstrate the importance of selecting appropriate element types and accurately replicating real-world boundary conditions when predicting the structural behavior of the 3D-printed truss system.

Conclusion

This study evaluated the structural performance of a 3D-printed truss member by comparing analytical predictions, FEA simulations, and experimental failure testing. Overall, the results should show that the CBEAM elements demonstrate the beam elements more accurately and represent real world behavior better due to their ability to model bending and shear effects that may occur. However, our experiment results showed how much printing and errors, assembly imperfections, and testing variability can influence failure behavior and reduce agreement with purely theoretical models. The discrepancies across results highlight the limitations of assuming idealized properties and boundary conditions when modeling and experimenting with 3D printed members. Despite the limitations and large discrepancies, this project provided insight into how calculations, simulations, and physical testing are used in and comparable in structural design. Improvements for future work may include increasing printing precision and applying real-world boundary conditions and limitations to simulations and calculated predictions. These improvements could strengthen the correlations between simulation, experimental, and analytical results, and create more reliable and accurate results.

References

[1] “Acrylonitrile butadiene styrene (ABS) - properties,” Acrylonitrile Butadiene Styrene (ABS) - Properties [Online]. Available: <https://matmake.com/materials-data/acrylonitrile-butadiene-styrene-properties.html>. [Accessed: 02-Dec-2025].

Appendix

Cross sectional area for ABS

$\sigma_y = 4150 \text{ psi}$

$\nu = 0.35$

$FS = 1$

Axial loading

measured load: -6.02

$$\sigma = \frac{P}{A} \quad \sigma = \sigma_y \text{ because } FS = 1$$

$$A = \frac{P}{\sigma_y} \quad A = \frac{10.35 \text{ lbs}}{4150 \text{ psi}} = 0.00249 \text{ in}^2$$

Figure 7. Hand-Calculations for Cross Sectional Area Before Applying Correction Factor



**Michigan
Technological
University**

<https://mtu.instructure.com/>

Mechanical and Aerospace Engineering

TO : David Wallis

FROM : Addie Behling, Johnathan Kengel, Sierra Manriquez

DATE : 09/23/2025

SUBJECT: Practice Session Report- Reverse Engineering 1

This memo provides an analysis of the Neu Master Pneumatic Brad Nailer NRF50, detailing how the product is used, potential misuse scenarios, and user feedback from real reviews. Functions of the Brad Nailer are analyzed through black-box modeling and a hypothesized function structure. The goal is to assess performance, limitations, and propose potential improvements to influence future design iterations.

Customers value the Brad Nailer's lightweight design and ease of use. However, potential misuse includes using the tool as a weapon or using it for tasks too light for its intended purpose. It is hypothesized that the Brad Nailer operates by converting potential energy from compressed air into kinetic energy to drive the nails.

The Neu Master Pneumatic Brad Nailer NRF50 is a lightweight, user-friendly tool well suited for its intended applications, though it carries risks if misused. It operates by converting kinetic energy into potential energy.

MEEM2901

Mechanical Engineering Practice Course – I

Practice Session Report- Reverse Engineering 1

Date: 09/23/2025

Lab Section: Thursday, 3:00

Student name: Addie Behling, Johnathan Kengel, Sierra Manriquez

Introduction

The Neu Master Pneumatic Brad Nailer NRF50 is an air-powered brad nail gun kit designed for both professional and DIY use. The kit features the Brad Nailer itself, a ¼ in. Industrial Plug (¼ in. NPT), two Allen Wrenches, a Bottle of Tool Oil, and a Blow-Molded Carry Case for convenient storage. Key features on the nail gun include an adjustable exhaust to change the direction of airflow away from the user and adjustable depth control for driving nails deep and flush to the work surface. Price around \$50, the Brad Nailer kit is advertised for multiple applications including heavy duty and light duty projects. The Brad Nailer kit is intended for those who are working on projects requiring nails such as roofing, flooring, shelving, upholstery installation, and DIY projects.



Figure 1. Unboxed Neu Master Pneumatic Brad Nailer NRF50 Components.

User Research

Many users have found this pneumatic nail gun to be easy and practical to use. The setup for this tool is fairly simple. For safety, the nails should be loaded into the magazine before connecting to an air compressor. An air compressor of less than 200 psi is required for this tool to work properly. Once the user has hooked up the nail gun to an air compressor and adjusted the direction of nails, the tool is ready to be used. While in use the user may experience some problems that they need to troubleshoot. These may include the nail jamming, or the user runs out of nails. Figure 2 shows a use case flowchart of how a user may operate this product.

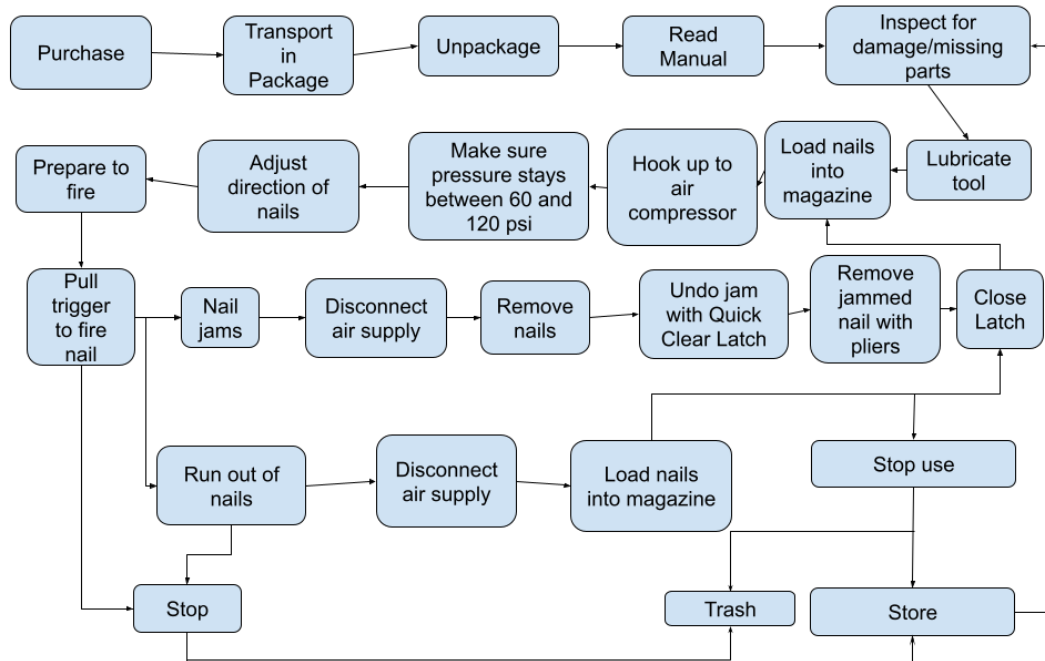


Figure 2. Use Case Step by Step Flowchart [1]

When using a pneumatic nail gun, there are many possible ways that a person may misuse this tool. Since this tool projects nails at a high velocity, misusing the nail gun could have very severe consequences. Using it in a violent way; as a weapon or aimed at a person, can cause severe injury and potentially death. There could also be severe consequences if the tool was used in the wrong setting. For example, this nail gun should never be used under water or in explosive atmospheres. This could cause severe damage to the tool, the surrounds, and the operator. Finally, this tool could be misused if it were to be utilized on minor projects, such as stapling paper, rather than putting nails into wood. This misuse could also cause damage to the tool, surroundings, and operator.

When this tool is used properly, costumers have had many positive experiences. Many people described the tool as being very light and easy to use [2], as well as that it works flawlessly and works great for the inexpensive cost [3]. Although this product seems to be highly rated across many retail platforms, there have been negative reviews and comments about this tool. There have many reviews stating that the box was busted upon arrival, the nail gun still leaves marks as if they were staples, the accessories are a terrible quality, and it double shoots/misfires occasionally. Although some of these may be the results of user error, one of the most received complaints has been about the poor quality of the accessories, which the users are unable to control. Overall, consumers seem to enjoy this product, saying it works well for household projects, is easy to use, and is relatively inexpensive.

Reverse Engineering and Hypotheses

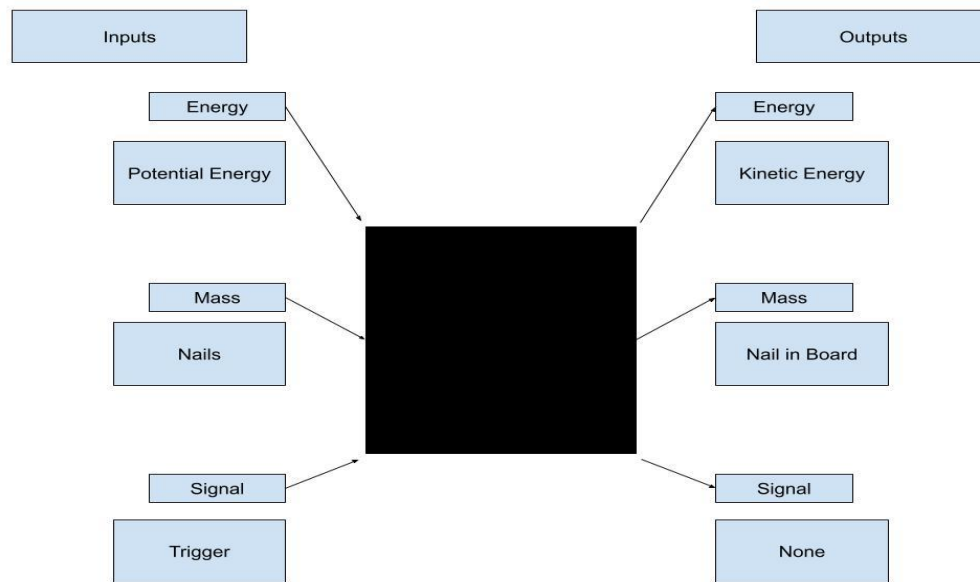


Figure 3. The Black Box Model

Figure 3 shows the various inputs and outputs of the system. The three different kinds of inputs each have their own effect on the system. The nail gun uses compressed air to fire a nail, as shown by figure 3 this compressed air is potential energy. The nail gun is loaded with nails which are mass. The nail gun uses the trigger as the signal to activate. When the nail gun activates it shoots the nail into the board, and releases air. The release of the nail is mass output; while the release and movement of air is kinetic energy output.

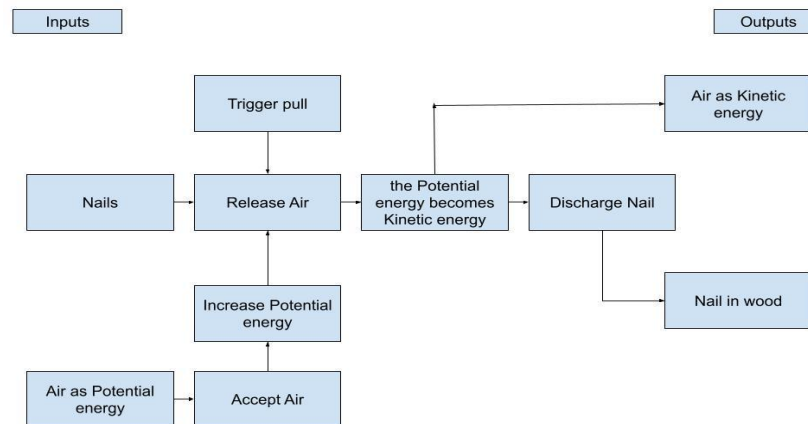


Figure 4. The Hypothesized Function Diagram

Figure 4 is the hypothesized interworking of the nail gun. As presented in figure 3 the inputs and outputs are the same. Figure 4 shows that air goes in where is used as potential energy. When the nail is in and the trigger is pulled the system activates releasing air. The released of the air

converts the potential energy into kinetic energy. The kinetic energy is used to launch the nail out and into a board.

Based on the research and hypothesis of the function we developed improvements and requirements for the Nail gun. The customers want a lightweight item so the nail guns weight will be less than five pounds. The costumers also want something easy to use so we will reduce the amount of user input required. The nail gun needs to be able to work with high enough power up to at least sixty psi. The nail gun is also required to be used within most households easily so, it will have a range of 60-120 psi. Lastly the nail gun must not leave marks on the product so it should be able to launch the nail through the board with the nozzle five centimeters from the board.

Conclusion

The analysis of the Neu Master Pneumatic Brad Nailer NRF50 highlights its lightweight design, ease of use, and effectiveness for a variety of different projects, while also identifying potential misuse risks. The hypothesized black-box model suggests that the system operates with three inputs: compressed air (energy), nails (mass), and the trigger (signal). These inputs produce outputs: driven nail, released air, energy transfer. Next steps should involve reverse engineering the Brad Nailer to confirm the hypothesized models to identify opportunities for performance improvements and safer operation.

References

- [1] Neu Master Pneumatic Brad Nailer NRF50 User Manual
- [2] Rajeania, March 15, 2025, Walmart Product Reviews [Online]. Available: <https://www.walmart.com/reviews/product/1515241266?ratings=5>.
- [3] Kathy, August 27, 2025, “Amazon Product Review,” Amazon.com [Online]

Appendix

Reverse Engineering 1

Positive User Experiences	Negative User Experiences
very light and easy to use	Box based on arrival
intuitive	when using note the gun still leaves
works flawlessly and well	marks as if it were a single
	too easily disconnects
	double sprays
	can adjust, glue, multi, and styles

amazon, walmart, ebay

Reverse Engineering 1 4

Figure 5. User Experiences

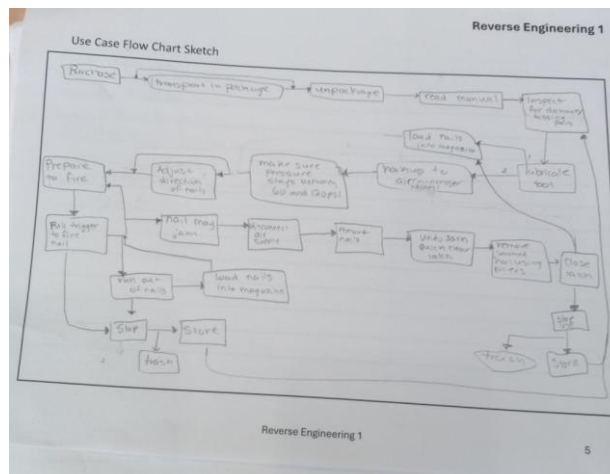


Figure 6. Use Case Flow Chart Sketch

Reverse Engineering 1

Possible misuse cases

- Aim at person
- Used under water
- Used to launch projectiles
- Used as weapon
- Used for sniper/longer term
- Used in explosive atmospheres

Reverse Engineering 1 6

Figure 7. Possible Misuse Cases

[illegible]

Figure 8. Customer Needs and Wants

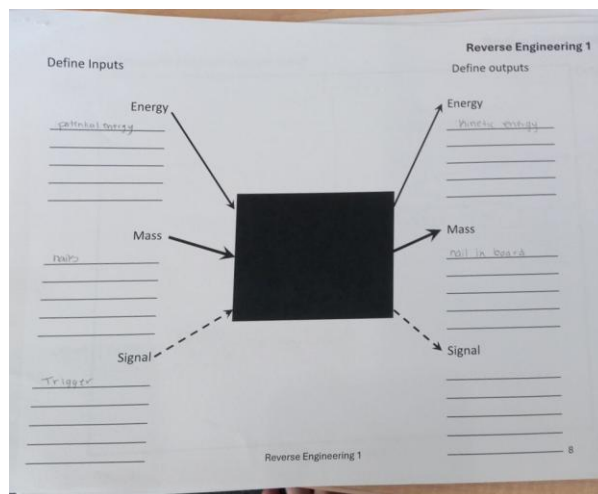


Figure 9. Black Box Chart

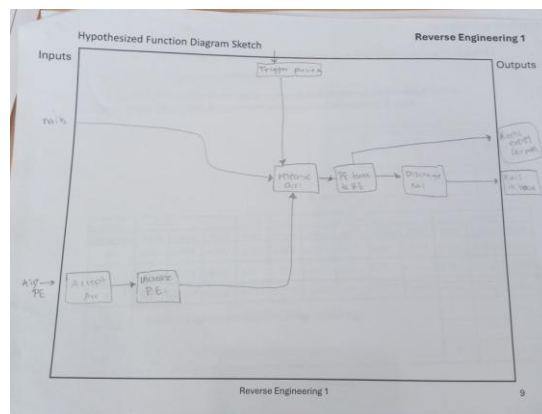


Figure 10. Hypothesized Function Diagram Sketch



Interoffice Memorandum

TO : David Wallis
FROM: Sierra Manriquez
DATE : 10/22/2025

SUBJECT: System Simulation

The purpose of this memo is to summarize the findings from the AMESim spring mass damper system model simulation and to show how system parameters impact the behavior of the system. This memo is intended to show the usefulness of AMESim for modeling systems and to support using simulation software in the engineering design process.

During this simulation, a spring mass damper system made of two spring dampers, and one mass was modeled in AMESim. The results showed that increasing the damper rating parameter on the spring from 19 N/(m/s) to 38 N/(m/s) cut the oscillation decay time in half from approximately 4 seconds to 2 seconds. This confirmed that there was a correlation between damping and the rate at which the system returns to equilibrium

The main takeaway from this lab is that AMESim demonstrates how damping impacts a spring mass damper system by dissipating the energy. The higher the damping value, the quicker the mass reaches equilibrium. These results show that AMESim is useful for analyzing systems and manipulating parameters.

MEEM2901

Mechanical Engineering Practice Course – I

System Simulation with AMESim

Date: 22-10-2025

Lab Section: Thursday, 03:00

Student name: Sierra Manriquez

Introduction

AMESim is a simulation software that is used to model and analyze systems. AMESim is used for engineering simulations to model, analyze, and optimize the performance of systems involving multiple components. The objective of this lab is to familiarize students with AMESim and its functions so that they have a base knowledge of how to use the software. Students are expected to be able to model a spring mass damper system, which is made up of two springs with one mass between the two springs.

Model Description

The purpose of the experiment is to let students learn how to use simulation tools in the engineering design process, how to use the basics of AMESim software, and how to analyze changes in system behavior. This experiment is important because it shows students the importance of using simulation software for testing and optimizing mechanical systems. For future projects, students must know the importance of using simulation software as part of the engineering design process before jumping straight into prototype testing in order to check calculations, simulate how the system will function, and catch any possible changes that need to be made before resources are used on a physical model.

The spring mass damper system used in this lab has three main components: mass, spring, and damper. The mass component is the part of the system that represents an object with mass that moves due to the forces from the springs. The spring component is the part of the system that provides a force proportional to the displacement of the mass. The spring component is described by Equation 1, Hooke's Law, which models the behavior of a spring where F is the force applied by the spring, k is the spring constant, and x is the displacement of the spring [1].

$$F = kx \quad (1)$$

The damper component is the part of the system that provides resistance to motion by dissipating the energy. The damper is described by Equation 2, which models the behavior of the damper where F is the damping force, c is the damping coefficient, and V is the velocity of the mass [1].

$$F = cV \quad (2)$$

The three main components act together as the spring mass damper system. Figure 1 shows the assembled spring mass damper system with each component labeled.

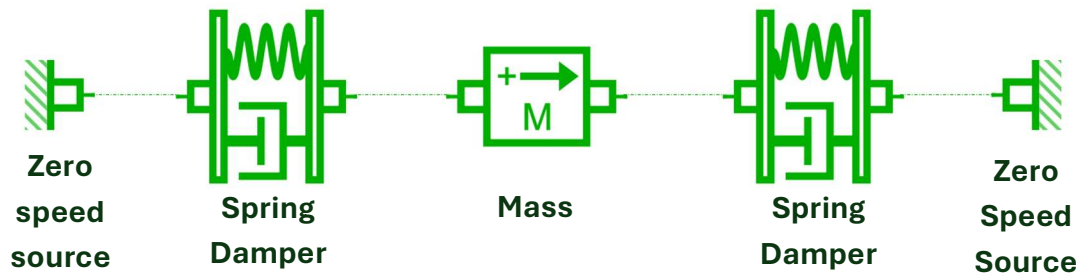


Figure 1. Labeled AMESim Spring Mass Damper System Model Diagram

In AMESim, the mass component functions as the body whose motion is being simulated with the influence of force from the spring and the damper. In this specific model, the parameters of the mass component are displacement and mass, which are equal to 0.1 m and 11 kg, respectively.

The two spring components model the elastic behavior of the system. The springs provide the restoring force that opposes the displacement of the mass component, making it return to its equilibrium position. In this specific model, the parameters of the spring components are spring rate and damper rating, which are equal to 20040 N/m and 0 N/m, respectively, and are identical in each spring. The damper component in the AMESim model of the spring mass damper system models energy loss and reduces oscillation of the mass component. In this system, the damper is connected to the spring as the damper rating parameter.

The zero speed source is used to create a fixed reference point for the system. The zero speed source is connected to either side of the system and has zero velocity, meaning that it does not move even if there are forces applied to it. In this system, the parameter of the zero speed source is linear displacement, which is equal to 0 m and is identical in both zero speed sources.

Changes to System Behavior

The damper element in AMESim is introduced into the spring mass damper model by configuring the damper rating parameter on the spring element. This can be done by selecting the spring element and changing the value to a positive non-zero value. For this specific system, the damper rating parameter was set to 19 N/(m/s). Figure 2 presents the simulation results by graphing the displacement of the mass element over time with damping.

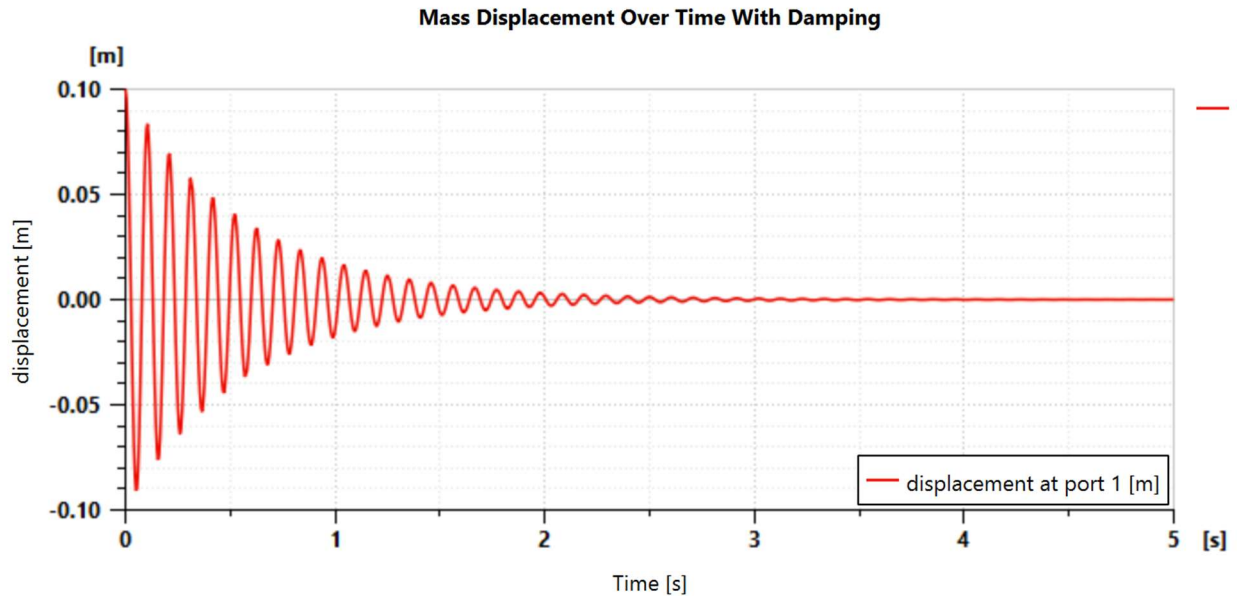


Figure 2. Plot of Displacement of the Mass Over Time with Damping

The decay in time is the amount of time until the amplitude is below 0.1% of the initial deflection. 0.1% of the initial deflection is calculated by taking the initial displacement, 0.1 m in this case, and multiplying it by 0.001 to get the desired threshold value. 0.1% of the initial deflection for this specific model is 0.0001 m, and the decay time is about 4 seconds. Figure 3 presents a zoomed in plot of the mass displacement over time at the point at which oscillation decays to 0.0001 m at 4 seconds; the blue dashed line represents the specific threshold of 0.0001 m [1].

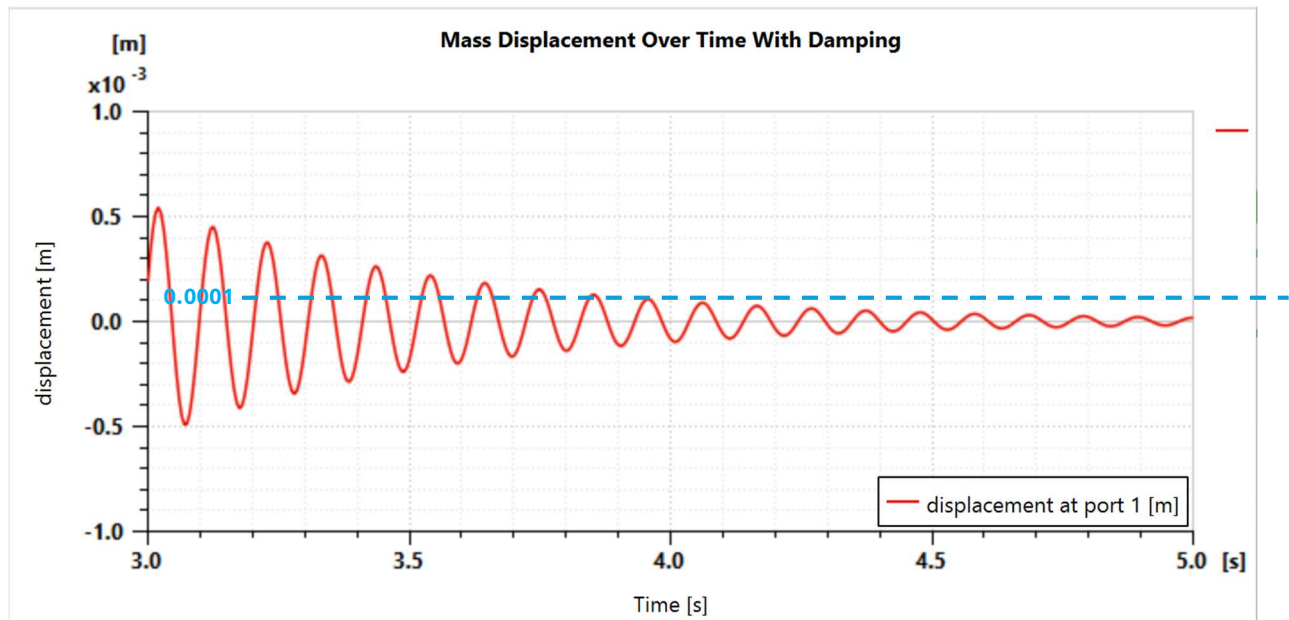


Figure 3. Zoomed in Mass Displacement over Time with Damping Plot where Oscillation Threshold is Reached

Increased Damping

The damper rating parameter can be further modified to cause the displacement of the mass to decay to zero in half the time it did before. By increasing the damper rating parameter to double its original value, the time taken for the mass displacement to reach less than 0.1% of the starting displacement is halved. For this model, the original damper rating was 19 N/(m/s), doubling this value gives a damper rating of 38 N/(m/s). The time taken for the displacement to reach the threshold is about 2 seconds which is half of the original time of 4 seconds. Figure 4 shows the plot of the displacement of the mass over time with increased damping.

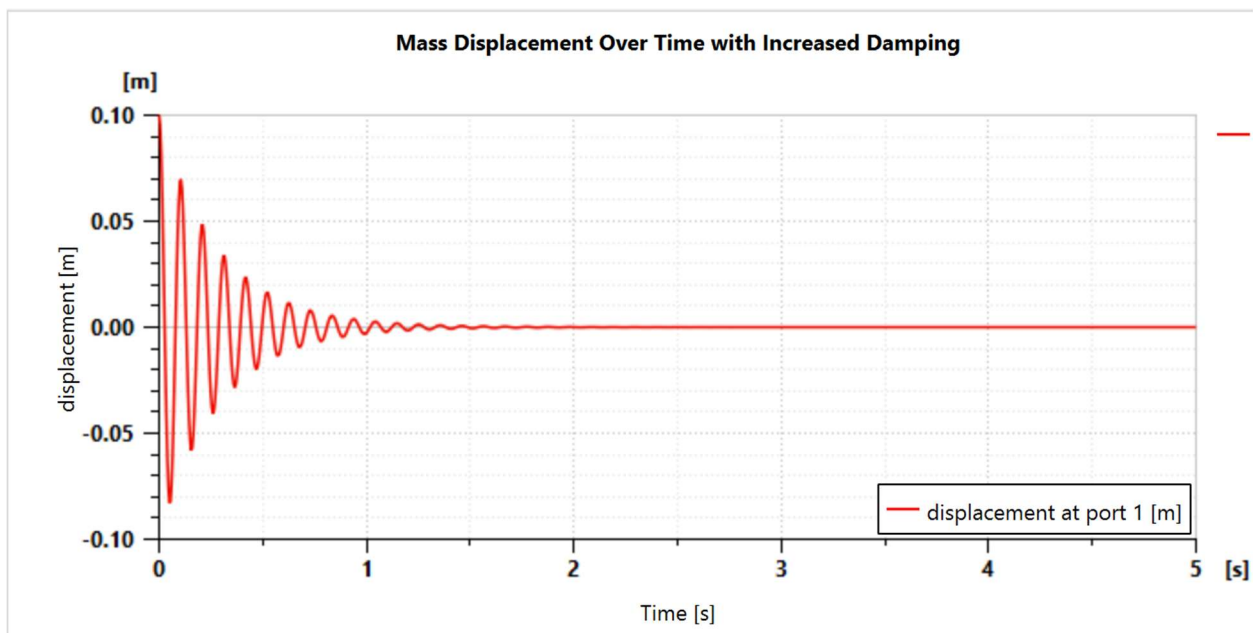


Figure 4. Plot of Displacement of the Mass Over Time with Increased Damping

Figure 5 presents the zoomed in plot of the mass displacement over time with the increased damping parameter. The blue dashed line shows the specific threshold of 0.0001 m.

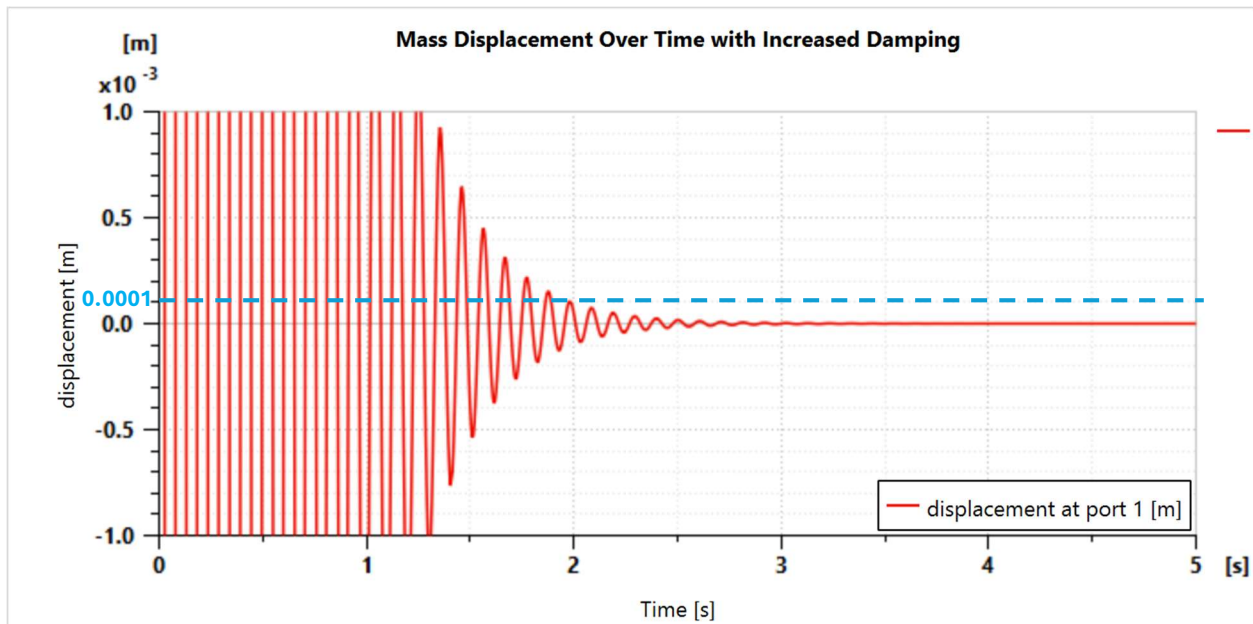


Figure 5. Zoomed in Plot of Displacement of the Mass Over Time with Increased Damping

Conclusions and Recommendations

The spring mass damper system in AMESim demonstrated how damping impacts system behavior, particularly the rate at which oscillations decay. The key finding from this simulation was the correlation between damping and decay time. This relationship showed that doubling the damper rating halved the decay time. This showed that the higher the damping rate parameter, the more energy dissipated, allowing the system to return to an equilibrium state in less time.

AMESim was an effective tool for modeling and simulating systems. The user interface was intuitive, and manipulating parameters and the system could be simulated quickly. This makes AMESim highly useful for understanding how a system will work with its individual components and what changes to parameters may be needed to increase efficiency. A limitation of AMESim is that the system could potentially oversimplify real-world behavior such as external disturbances.

For future use or similar projects, it is recommended that the parameters for the components are chosen carefully so that the simulation behaves how the physical model would in real life. The parameter values can be chosen based on measurements from real components, which would increase the accuracy of the simulation.

References (Word Style: Heading 1)

- [1] "Amesim Practice Session Instructions", in ME2901 canvas course, last accessed on 10/21/2025
- [2] "Lecture8A – Amesim Modeling", in ME2901 canvas course, last accessed on 10/21/2025